Master of Science in Computer Science
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# Artificial Intelligence

## Driven Optimization Framework for Construction Site Ecosystems

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The authors declare that they are the sole authors of this thesis and that they have not used any sources other than those listed in the bibliography and identified as references. They further declare that they have not submitted this thesis at any other institution to obtain a degree.


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# Abstract

**Background:** Construction sites are complex ecosystems characterized by dynamic workflows, heterogeneous machinery, and diverse stakeholders. Traditional site management often suffers from inefficiencies, cost overruns, and sustainability challenges due to fragmented decision-making and limited use of real-time data. Emerging advances in Artificial Intelligence (AI), predictive analytics, and digital optimization provide new opportunities to transform construction site operations into integrated, intelligent and sustainable ecosystems.

**Objectives:** This thesis proposes and evaluates an *AI-Driven Optimization Framework* that addresses critical challenges in construction site ecosystems. The framework is designed to enhance operational efficiency through intelligent scheduling, resource allocation, and passage guidance; reduce downtime and costs by enabling predictive maintenance of multi-brand machinery; improve sustainability by minimizing fuel consumption, carbon emissions and material waste; and ensure scalability and interoperability across diverse construction contexts.

**Methods:** A *Combined Formal Experiment and Case Methodology* was adopted, combining reinforcement learning, predictive modeling, and simulation-based evaluation. Real-world construction datasets, IoT sensor inputs, and synthetic scenarios were utilized to train and validate the framework. Performance was assessed using established metrics such as task completion time, resource utilization, maintenance efficiency, and environmental impact.

**Results:** The framework achieved significant improvements across key dimensions. Predictive maintenance extended equipment life and reduced unplanned failures. AI-based passage guidance minimized travel distances, leading to lower emissions and fuel consumption. Intelligent scheduling enhanced coordination among workers and equipment, thereby improving productivity. Pilot validations confirmed the framework's adaptability to diverse construction settings while demonstrating measurable cost savings and sustainability gains compared to conventional approaches.

**Conclusions:** The study confirms the feasibility and effectiveness of integrating AI-driven optimization in construction site ecosystems. By unifying predictive, prescriptive, and sustainable decision-making techniques, the framework not only enhances operational efficiency but also contributes to safer and more environmentally responsible construction practices. This research advances the digital transformation of the construction industry and lays the foundation for broader industrial adoption of AI-powered optimization frameworks.



**Keywords:** Artificial Intelligence, Optimization Framework, Construction Site Ecosystems, Predictive Maintenance, Sustainable Construction

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# Abbreviations

| | | |
|---|---|---|
| **AI** | : | Artificial Intelligence |
| **AR** | : | Augmented Reality |
| **COT** | : | Chain of Thought |
| **CV** | : | Computer Vision |
| **FAISS** | : | Facebook AI Similarity Search |
| **F1** | : | F1-Score (harmonic mean of precision and recall) |
| **FMS** | : | Fleet Management System |
| **IoT** | : | Internet of Things |
| **LLM** | : | Large Language Model |
| **ML** | : | Machine Learning |
| **MTTR** | : | Mean Time to Resolution |
| **NLP** | : | Natural Language Processing |
| **PANNs** | : | Pretrained Audio Neural Networks |
| **PROSIS** | : | Product Support Information System |
| **RAG** | : | Retrieval Augmented Generation |
| **ROI** | : | Return on Investment |
| **UI** | : | User Interface |
| **YOLOv8** | : | You Only Look Once, Version 8 (object detection model) |

# Chapter 1

# Introduction

Construction projects are inherently complex undertakings, involving heavy machinery, skilled labor, and tightly coupled workflows that must operate seamlessly to meet demanding schedules. Among the many challenges faced by the industry, unplanned equipment downtime remains one of the most disruptive and costly factors. When machinery fails, productivity is immediately compromised, operations are brought to a halt, and project costs escalate sharply. This challenge becomes particularly critical in remote sites, such as quarries, where immediate access to technical expertise is often unavailable. Even relatively minor mechanical problems, such as clogged filters, loose hoses, or faulty sensors, can trigger significant delays when operators lack the necessary tools or knowledge to address them on-site.

Empirical studies indicate that more than half of the equipment faults leading to downtime are minor issues that could typically be resolved within minutes [22]. In practice, however, these faults frequently result in prolonged interruptions, as operations are forced to wait for technicians to arrive and perform diagnostics. In Latin America alone, such delays are estimated to cause more than four million lost work hours annually, amounting to financial losses exceeding $15 million. At the global level, unplanned downtime in construction is reported to cost companies up to $50 billion each year, with individual incidents generating losses between $22,000 and $50,000 per hour [37, 58]. These figures underscore the substantial economic and operational burden associated with downtime in construction environments.

To mitigate these inefficiencies, recent research initiatives have explored the application of Artificial Intelligence (AI) as a means to improve efficiency and reduce downtime on construction sites [8]. The original scope of such projects has often been broad, envisioning AI-based frameworks capable of optimizing diverse aspects of site operations, including predictive maintenance, resource allocation, and machine routing [46, 60]. However, insights gained through field studies and interviews with operators, managers, dealers, and technicians have revealed a more pressing challenge. While operators frequently recognize that a problem exists, they often lack the confidence, guidance, or technical support to resolve it independently.

Building on this insight, research efforts have increasingly shifted toward empowering operators at the point of need. Rather than remaining dependent on technician intervention, operators can be supported with intelligent tools that enable them to act promptly and safely. The proposed solution is a Conversational AI agent, designed not as a simple FAQ system but as an advanced assistant capable of providing step-by-step guidance for troubleshooting and repair [24]. By leveraging state-of-the-art AI capabilities within digital support ecosystems, the agent seeks to transform downtime into productive action time, thereby reducing unnecessary technician dispatches and mitigating associated costs.

The proposed Conversational AI agent distinguishes itself through its integration of multiple AI technologies within a single, user-oriented system. It employs natural language processing to interpret operator queries, computer vision to analyze images of machine components and damages [51, 62], and retrieval-augmented generation to access contextually relevant information from product support databases and technical manuals [16, 30]. In addition to text-based queries, the system accepts multimodal inputs such as images and audio recordings, allowing operators to upload photographs of damaged components or submit audio files of abnormal machine sounds [4, 28]. This capability enables the agent to deliver accurate and context-specific diagnostic support tailored to real-world conditions.

By shifting maintenance support from a reactive to a proactive paradigm, the Conversational AI agent simultaneously addresses two key challenges: reducing costly downtime and capturing valuable field data. Operators are enabled to address minor issues before they escalate, while organizations benefit from insights into common failures, operational patterns, and machine performance [9]. These insights can then inform iterative improvements in product design, service strategies, and predictive maintenance practices. As such, the Conversational AI agent functions not only as a problem-solving tool but also as a mechanism for knowledge transfer between field operations and organizational learning.

The objective of this thesis is to design, implement, and evaluate a Conversational AI agent as a proof-of-concept Product-Service System (PSS). The work adopts a human-centered design methodology to ensure alignment with operator requirements and practical constraints [2]. The technical implementation incorporates a YOLOv8-based vision module for fault detection [62] and a Retrieval-Augmented Generation module for knowledge retrieval [30]. The evaluation is conducted using simulated field scenarios to examine both system performance and usability [14]. Through this approach, the thesis aims to demonstrate the potential of AI-driven tools to support operators in reducing downtime and improving site-level efficiency.

Beyond the immediate scope of this project, the work contributes to a broader digital transformation currently underway within the construction industry. Despite being one of the largest economic sectors globally, construction remains among the least digitized. Maintenance practices are still largely reactive, with interventions initiated only after failures occur [34]. Combined with the increasing complexity of modern machinery and a shortage of skilled technicians, this reactive approach perpetuates inefficiencies. The integration of AI technologies offers a promising pathway out of this cycle. By enabling predictive analytics, decision support, and interactive assistance, AI has the potential to deliver cost savings of 10-15% across construction projects [11].

Modern construction sites generate vast amounts of data from sensors, IoT devices, and operational systems. However, much of this data remains underutilized due to fragmented infrastructures and insufficient intelligent processing. The Conversational AI agent concept demonstrates how these data-rich environments can be leveraged to create smarter, more resilient operations. Positioned at the intersection of human operators and digital intelligence, the agent redefines on-site support by shifting from external dependency to immediate, guided action. In doing so, this thesis not only introduces a practical proof-of-concept system but also illustrates the broader potential of AI to reshape maintenance workflows and unlock efficiency gains across the construction industry.

## 1.1 Problem Statement

Construction sites are dynamic environments where even small inefficiencies can escalate into significant delays, safety risks, and cost overruns. While emerging AI-driven solutions show promise in streamlining operations and optimizing equipment performance [56], a critical challenge remains the lack of interoperability across mixed fleets from different manufacturers. To achieve true site-wide efficiency, an AI solution must be brand-agnostic, integrating excavators, trucks, conveyors, and human operators into a unified data ecosystem.

This thesis specifically addresses the operational bottleneck of unplanned downtime caused by minor equipment faults. The current reactive, technician-dependent model for troubleshooting is inefficient, leading to prolonged interruptions [39]. A paradigm shift is needed towards proactive, operator-enabled problem resolution. Therefore, the core problem this research tackles is the absence of an intelligent integrated support system that empowers operators to perform immediate, informed diagnostics and initial troubleshooting, thereby reducing dependency on external technicians and minimizing disruptive downtime.

## 1.2 Aim

This thesis aims to develop and evaluate a Conversational AI agent system that transforms construction site troubleshooting workflows by embedding AI-powered conversational interfaces into operations, thereby redefining maintenance response strategies from reactive approaches to proactive, operator-enabled problem resolution.

## 1.3 Objectives

To fulfill the stated aim, this thesis is guided by the following specific objectives:

1. **Develop a Conversational AI Agent Framework:** Design and implement a multimodal AI system capable of processing text, image, and audio inputs to

provide step-by-step troubleshooting guidance for common construction equipment issues.

2. **Evaluate Troubleshooting Speed and Efficiency:** Conduct controlled experiments to quantify the impact of AI-assisted workflows on time-to-resolution, time-to-first-action, and overall equipment downtime compared to traditional technician-dependent approaches.

3. **Assess Technician Intervention Reduction:** Analyze the effectiveness of the AI agent in enabling operators to resolve routine maintenance issues independently, measuring the reduction in unnecessary technician dispatches through comprehensive case study documentation.

4. **Improve Communication Quality:** Establish standardized, structured communication protocols between operators, technicians, and management teams through automated service ticket generation and tracking.

5. **Establish Performance Benchmarks:** Create measurable metrics for evaluating AI-assisted troubleshooting effectiveness, including first-time resolution rates, operator confidence levels, and overall impact on construction site productivity.

### 1.3.1 Research Questions

The research is driven by the following questions, designed to address the core problem statement and achieve the stated objectives:

**RQ1: How can embedding a Conversational AI Agent into construction workflows redefine troubleshooting speed and reduce downtime compared to technician-dependent approaches?**

***Research Methodology:*** *Formal Experiment*
***Justification:*** *An experimental approach is justified because it enables a controlled comparison between technician-dependent and AI-assisted workflows. By simulating common machine faults and measuring key metrics like time-to-diagnosis, time-to-resolution, and downtime, the study can isolate the AI agent's effect and provide statistically rigorous evidence of its efficiency gains.*

**RQ2: What role can a Conversational AI Agent play in minimizing unnecessary technician interventions for routine machine issues at construction sites?**

***Research Methodology:*** *Case Study*
***Justification:*** *A case study methodology is justified to rigorously analyze how the AI agent supports operator-led troubleshooting and reduces dependence on technician intervention within construction site workflows. By combining quantitative indicators with qualitative operator insights, this approach provides a comprehensive and context-aware assessment of the agent's role in optimizing maintenance efficiency.*

### 1.3.2 Ethical, Societal, and Sustainability Considerations

The development and deployment of an AI system in a safety-critical environment like a construction site necessitates a rigorous examination of its broader implications. This research will be conducted with a strong commitment to ethical principles, societal benefit, and environmental sustainability.

**Ethical Aspects:** The primary ethical concerns involve data privacy and security, as the system may process sensitive operational data. Strict protocols for data anonymization and secure storage will be implemented. Algorithmic transparency and accountability are paramount; the system will be designed with explainable AI (XAI) principles to ensure operators understand its recommendations and retain final decision-making authority [2, 29]. The research will actively mitigate risks of algorithmic bias and prevent the deskilling of operators by positioning the AI as an assistive tool, not a replacement for human expertise [10, 13].

**Societal Aspects:** The societal impact is potentially significant. By enabling faster resolution of equipment issues, the system aims to reduce accident risks associated with malfunctions and improper repairs, directly contributing to improved construction site safety [15, 27]. It also addresses the critical skilled labor shortage by democratizing technical knowledge, allowing less experienced operators to perform complex tasks with expert guidance, thus preserving institutional knowledge and creating new career pathways [8].

**Sustainability Aspects:** From an environmental perspective, the AI agent promotes sustainability by extending equipment lifespan through improved maintenance, reducing the resource footprint of manufacturing new machinery [46]. Minimizing unnecessary technician dispatches directly lowers carbon emissions from service vehicles. Furthermore, by ensuring equipment operates at optimal performance levels, the system contributes to reduced fuel consumption and lower emissions during construction operations [9, 68]. The research will quantify these environmental benefits where possible.

## 1.4 Outline

This thesis is organized into the following chapters:

- **Introduction:** Provides an overview of the research problem, aim, and objectives. It introduces the motivation for developing an AI-driven Virtual Assistant for construction equipment maintenance and sets the scope of the study.

- **Background:** Lays the foundations for key concepts relevant to this research, including AI in construction (Construction 4.0), multimodal learning, LLMs, RAG, and CV (YOLOv8). It also explains how these technologies intersect with maintenance workflows

  .

- **Related Work:** Highlights existing literature on AI applications in industrial maintenance and construction. Identifies gaps in current approaches, particularly the lack of integrated multimodal systems and operator-focused solutions.
- **Methodology:** Explains the research design, including preprocessing of knowledge sources, embedding generation with FAISS, multimodal data handling, experimental setup, and evaluation metrics. Describes how both simulated and field test scenarios were carried out.
- **Results and Analysis:** Presents the quantitative and qualitative results of the experiments, such as accuracy rates of NLP and CV modules, MTTR reduction, technician dispatch reduction, operator resolution rates, and economic benefits (ROI). Interprets these findings with respect to the research questions.
- **Discussion:** Interprets the significance of results, focusing on accuracy, usability, human-AI collaboration, safety considerations, organizational impact, and cultural acceptance. Discusses limitations such as dependency on connectivity, knowledge base coverage, and environmental sensitivity of CV models.
- **Conclusion and Future Work:** Summarizes the key findings, emphasizing contributions such as operator empowerment, downtime reduction, and knowledge democratization. Outlines directions for future research and development, including predictive maintenance integration, advanced multimodal perception (audio and thermal), offline deployment, and the development of both mobile and remote applications for the AI assistant.

# Chapter 2

# Background

## 2.1 AI Chatbots in Maintenance and Field Service

The use of AI-driven conversational agents for technical support has grown rapidly across industries. Maintenance operations benefit from AI chatbots that can leverage large knowledge bases to troubleshoot issues [8, 24, 56]. Reports indicate that modern AI chat systems can analyze symptoms and instantly pull solutions from large databases of maintenance logs, repair manuals, and expert knowledge [24]. By doing so, these chatbots automate what was traditionally a human expert's role, provide immediate guidance, and reduce the need for initial expert intervention [8, 24].

| **Study** | **Chatbot Type** | **Primary User** | **Knowledge Source** | **Reported Benefits / Notes** |
|---|---|---|---|---|
| Johnson et al. (2021) [24] | Hybrid (rule-based + retrieval/dialog) | Technicians | Technical docs / KB (enterprise) | Faster access to diagnostic guidance; reduced back-and-forth communications; improved support process usability (design + evaluation). |
| Chen et al. (2024) [8] | Survey / Mixed (rule-based, hybrid, LLM) | Technicians / Operators (varies) | Manuals, logs, sensor data (varies) | Consolidates evidence of AI benefits in equipment maintenance; reports improvements in productivity, cost reduction, and decision support across deployments. |
| Saka et al. (2024) [56] | LLM / GPT-based assistant | Construction stakeholders | General + domain prompts (LLM-centric) | Validates GPT use cases in construction; highlights opportunities and limitations; indicates potential to reduce response latency and initial expert dependence. |

Table 2.1: Comparison of chatbot approaches in maintenance (representative studies).

### 2.1.1 Industry Implementations

CNH Industrial *parent of Case and New Holland brands* deployed an AI chatbot for dealer technicians that allows queries over approximately 1.5 million pages of technical manuals and returns precise diagnostic guidance. Trade coverage credits this AI Tech Assistant with speeding up repairs and improving first-time fix rates in dealer service centers. Our work adopts a similar concept but brings it directly to equipment operators at the jobsite. The goal is to extend AI support to the front lines and not only to technicians [8].

### 2.1.2 Research on Hybrid Chatbots

There is a growing body of research on hybrid AI assistants for maintenance. One study developed a custom chatbot that combines rule-based prompts with generative AI to streamline maintenance communications and reduce downtime [24]. The chatbot simulates a human-machine interface that guides maintenance staff, and early experiments indicated reduced buffer times in maintenance processes. Another research effort examined conversational AI for real-time equipment monitoring, where bots engage in dialogue with maintenance personnel to walk through troubleshooting procedures and also predict failures from conversational context [4, 40]. These efforts underline the value of natural and contextual interaction in industrial settings. We incorporate this aspect through an intuitive chat interface and multi-turn dialogue design. Our AI agent differs by tightly integrating retrieval for factual accuracy and multimodal inputs. It can handle text descriptions and images today, and it is designed to incorporate audio in future work [28].

| | Challenges | | | | |
|---|---|---|---|---|---|
| Applications | Representation | Translation | Alignment | Fusion | Co-learning |
| **Speech recognition** | | | | | |
| Audio-visual speech recognition | ✓ | | ✓ | ✓ | ✓ |
| **Event detection** | | | | | |
| Action classification | ✓ | | | ✓ | ✓ |
| Multimedia event detection | ✓ | | | ✓ | ✓ |
| **Emotion and affect** | | | | | |
| Recognition | ✓ | | ✓ | ✓ | ✓ |
| Synthesis | ✓ | ✓ | | | |
| **Media description** | | | | | |
| Image description | ✓ | ✓ | ✓ | | ✓ |
| Video description | ✓ | ✓ | ✓ | ✓ | ✓ |
| Visual question-answering | ✓ | | ✓ | ✓ | ✓ |
| Media summarization | ✓ | ✓ | | ✓ | |
| **Multimedia retrieval** | | | | | |
| Cross modal retrieval | ✓ | ✓ | ✓ | | ✓ |
| Cross modal hashing | ✓ | | | | ✓ |
| **Multimedia generation** | | | | | |
| (Visual) speech and sound synthesis | ✓ | ✓ | | | |
| Image and scene generation | ✓ | ✓ | | | |

Figure 2.1: hybrid AI chatbot design [4]

## 2.2 Retrieval-Augmented Generation and Knowledge-Based NLP

Large Language Models have strong capabilities in understanding and generating text, but hallucinations and limited access to up-to-date or domain-specific information can be challenges. Retrieval-Augmented Generation *RAG* addresses these challenges by pairing the model with an external knowledge source that is queried during generation [16, 18, 30].

### 2.2.1 Foundational Work

The seminal RAG work for knowledge-intensive tasks combines a parametric neural model with a non-parametric memory *a searchable document index* [18, 30]. By retrieving relevant text snippets from resources such as Wikipedia or documentation and conditioning the language model on them, the system produces more accurate and verifiable outputs [17, 25, 53, 54].

### 2.2.2 Domain-Specific Adaptation

Our assistant builds on the RAG paradigm with a domain-specific knowledge base of construction machinery manuals and service bulletins. We use text-embedding models together with a FAISS index to enable semantic search over these documents [17, 23, 53]. This allows the assistant to quote exact troubleshooting steps or specifications from official manuals when responding to user queries. Such grounded responses are essential in maintenance to ensure accuracy and build user trust. We further organize the corpus into sub-indices per machine model and topic *error codes, troubleshooting guides, and similar* to improve relevance and speed. Retrieval orchestration is implemented with a production-grade framework *for example, LangChain*, yielding a robust pipeline that mitigates hallucinations by tying answers back to known sources [14, 16].

## 2.3 Computer Vision for Fault Detection

Visual inspection is central to equipment maintenance. Traditional methods rely on human observation to detect issues such as leaks, cracks, or abnormal wear. In recent years, computer vision models have automated fault detection in industrial settings [59, 67, 71].

### 2.3.1 YOLO for Real-Time Detection

Object detection networks from the YOLO family have been especially popular because they deliver real-time speed and strong accuracy [5, 51, 52, 62, 63]. YOLO reframes detection as a single end-to-end regression problem and achieves high frames per second while maintaining good mean Average Precision [32]. Since the original introduction in 2016, YOLO has evolved through multiple versions. The latest

YOLOv8 from 2023 demonstrates higher accuracy across varied domains compared to many predecessors [62].

### 2.3.2 Applications in Maintenance

In maintenance, researchers have trained vision systems to identify surface defects, corrosion, and leaks using YOLO variants or Faster R-CNN with promising results on controlled datasets [1, 15, 27, 69]. Our project contributes to this area by developing custom YOLOv8 models that detect construction machine component failures. We curated datasets for common issues. One dataset targets damaged wires *frayed or cut electrical wires*. Another dataset covers component failures such as corrosion, worn hoses, broken pistons, and fluid leaks [55]. These classes reflect typical failure modes in heavy machinery. Automated detection can enable predictive maintenance by catching problems early [22, 26, 60].

Deploying computer vision for continuous monitoring can reduce unplanned downtime by flagging wear and tear before catastrophic failure [**?**,37]. Although much academic work focuses on high-end industrial equipment and controlled environments, our results show that a lightweight vision model integrated into a maintenance assistant adds value by augmenting diagnostic workflows with visual analysis [67, 71]. When users share a photo, the assistant can identify visible symptoms *for example, a leaking hose versus a loose cap*. This provides context and confidence that pure text chat lacks.

## 2.4 Technician Scheduling and FMS

Optimizing the logistics of technical support is another lever for reducing downtime. Operations research has long studied the technician routing and scheduling problem. Traditional approaches assume accurate advance knowledge of failures. Predictive maintenance combined with dynamic scheduling improves outcomes [9, 46].

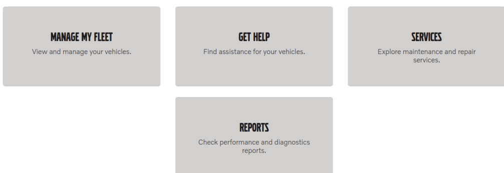



Figure 2.2: User Interface

### 2.4.1 AI for Field Service Optimization

Industry case studies report that incorporating AI into field service management allows dispatchers to optimize schedules and routes in real time [20, 68, 70]. This reduces travel time and increases job completion rates by matching technician skill, location, job priority, and traffic conditions to the job at hand. Field trials have shown that AI-based scheduling can reduce technician idle time and mean time to repair and can balance workload across teams [11]. Academic work has modeled multi-objective technician routing with skill constraints and demonstrated cost reductions using AI optimization [46].

### 2.4.2 Integration in Our Framework

Our framework addresses this space through technician and dealer dashboard modules. Inspired by prior work, we include features for technicians to set availability and for the system to recommend dispatch only when necessary. The AI assistant acts as a triage layer. By resolving a portion of issues remotely, it frees technician schedules. When a site visit is warranted, the system uses captured context *machine type, likely fault, parts needed* to streamline scheduling. This enables assignment to a technician with the right expertise and supports first-visit fix by recommending the correct spare parts [6, 36, 38, 43, 47]. Integrating scheduling with diagnostics advances beyond static scheduling tools. Field interviews emphasized that technicians often spend more time traveling than fixing because many calls are minor issues or false alarms. By handling those cases through the AI assistant, travel hours can be reduced. This improves efficiency and sustainability [39, 44].

## 2.5 Chain of Thought

Chain of Thought is a prompting technique for large language models in which the model generates intermediate reasoning steps before producing a final answer [64,65]. This improves accuracy and interpretability in complex multi-step tasks such as diagnostic reasoning. By structuring outputs into logical steps, CoT reduces errors and enhances transparency. In this thesis, CoT enables the Conversational AI agent to articulate its diagnostic reasoning, break down problem solving into sequential steps that operators can follow, and support verification in high-stakes construction environments [40, 48, 57].

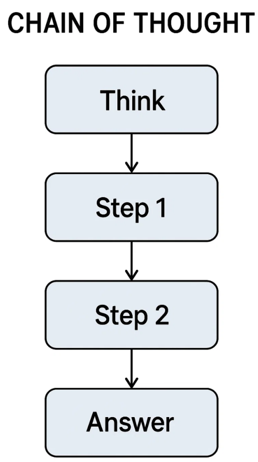



Figure 2.3: Chain of Thought

## 2.6 Roboflow

Roboflow is an end-to-end platform for computer vision workflows. It provides tools for dataset annotation, preprocessing, augmentation, and model deployment [55]. It integrates with popular architectures such as YOLO, which makes it useful for object detection systems [62]. In this thesis, Roboflow was used to curate and prepare datasets of construction machinery faults including hose damage, oil leaks, and coolant leaks. Field-collected images were augmented with synthetic variations to improve generalization. This streamlined process produced high-quality training data for YOLOv8 models and improved the robustness and accuracy of visual diagnostics.

To train the YOLOv8s models, we used two open-source Roboflow Universe datasets: Damage Wires and Component Failure Detection. The Damage Wires dataset (project by "Meine Arbeit") contains 428 annotated images across its v1 release; labels cover damaged electrical wiring in varied contexts. The Component Failure Detection dataset (project by "Nadia rjvo3") contains 752 annotated images in the referenced version, capturing a broader set of component-level faults (e.g., corrosion, wear, leaks, cracks). Together, these datasets provided 1,180 total images for supervised training of fault-detection models in construction-equipment scenarios.


Figure 2.4: Snippet of the trained dataset

## 2.7 Facebook AI Similarity Search

FAISS is an open-source library developed for efficient similarity search and clustering of dense vector embeddings. It is optimized for large-scale retrieval and supports indexing and querying billions of high-dimensional vectors in real time [23]. FAISS is widely used in RAG pipelines to quickly identify the most relevant documents or knowledge entries from large repositories. In this thesis, FAISS enables fast access to equipment manuals, service records, and prior fault histories so that troubleshooting recommendations are both contextually grounded and computationally efficient.

## 2.8 Product Support Information System

PROSIS is a digital platform for product support that provides access to service manuals, parts catalogs, and technical documentation. It centralizes authoritative information for maintenance and repair workflows so that technicians and operators can quickly locate accurate and manufacturer-verified data. In this thesis, PROSIS is integrated into the RAG pipeline. The result is troubleshooting advice grounded in official documentation, stronger technical accuracy, higher operator trust, and alignment with product support standards [8, 56].

**Poor steering response with lever steering**

Visar vald profil

| Gäller för serienummer | | | |
|---|---|---|---|
| **Modell** | **Tillverkningsort** | **Serienummer start** | **Serienummer stop** |
| L60H Volvo | | | |

**Description:**
The steering movement of the machine does not correspond to the lever movement.
Max steering speed is reached at a small lever movement.
The steering movement is rough with jerky start and stop.

**This decision tree assumes that the following has been checked and corrected:**

- Primary steering system is working
- No active error codes

**Possible causes:**

- Fault in the hydraulic system
- Fault in the electronic system
- Calibration error

Guided Diagnostics

Figure 2.5: Snippet of the PROSIS dataset

## 2.9 Pretrained Audio Neural Networks

Pretrained Audio Neural Networks are trained on large-scale audio datasets to extract general-purpose audio features [28]. They can recognize and classify a wide range of sound patterns without extensive task-specific training. In industrial contexts, PaNNs are useful for predictive maintenance because they detect abnormal machine sounds such as irregular vibrations, grinding, or engine anomalies that often precede equipment failure [22, 26]. In this research, PaNNs contribute to fault detection by providing the AI agent with an additional sensory modality and complement vision and text-based diagnostics with audio-driven anomaly detection.

## 2.10 Industry 4.0 and Maintenance Optimization

Leveraging advanced technology to optimize construction workflows aligns with the Industry 4.0 movement. This movement emphasizes connectivity, data analytics, and automation in traditionally manual industries [9, 46].

### 2.10.1 Digital Transformation in Construction

Digital transformation in construction has been slower than in manufacturing but is accelerating. Adopting digital workflows can improve efficiency. For example, mobile and cloud-based site tools can increase productivity and reduce costs on projects [8, 11].

### 2.10.2 Predictive Maintenance and FSM Systems

Predictive maintenance uses IoT sensors and machine learning to schedule repairs before failures occur [22, 33, 34, 60]. Research spans methods from logistic regression to deep neural networks that operate on sensor data to predict failures. These techniques complement our work. Predictive maintenance seeks to avert failures. Our AI assistant focuses on rapid response when failures occur. The assistant can also integrate predictive signals. If telemetry suggests an upcoming issue, the assistant can issue proactive warnings [39, 44].

Field Service Management systems optimize dispatching and service workflows. Many solutions incorporate AI for scheduling and augmented reality for remote assistance [6, 36, 38, 43, 47]. Remote assistance can eliminate unnecessary travel, cut carbon emissions, and accelerate problem resolution by enabling experts to help virtually. Our escalation feature aligns with this pattern. If the AI cannot solve an issue, it initiates a remote expert session by chat or video to avoid a truck roll when a remote solution is feasible.

### 2.10.3 Sustainability and Green Maintenance

Sustainability in maintenance is gaining attention under the term green maintenance. Strategies such as remote diagnostics, condition monitoring, and smarter scheduling can reduce emissions substantially [9, 46]. Our assistant supports these goals by enabling local problem solving and smart triage so that physical visits are reserved for cases where they are necessary and likely to succeed on the first attempt.

## 2.11 Standards and Interoperability

Integration with existing systems and standards is essential for an AI assistant in maintenance. Maintenance operations often follow frameworks such as ISO 14224 for equipment failure reporting and ISO 20815 for production assurance. These standards emphasize consistent data collection on failures and maintenance actions.

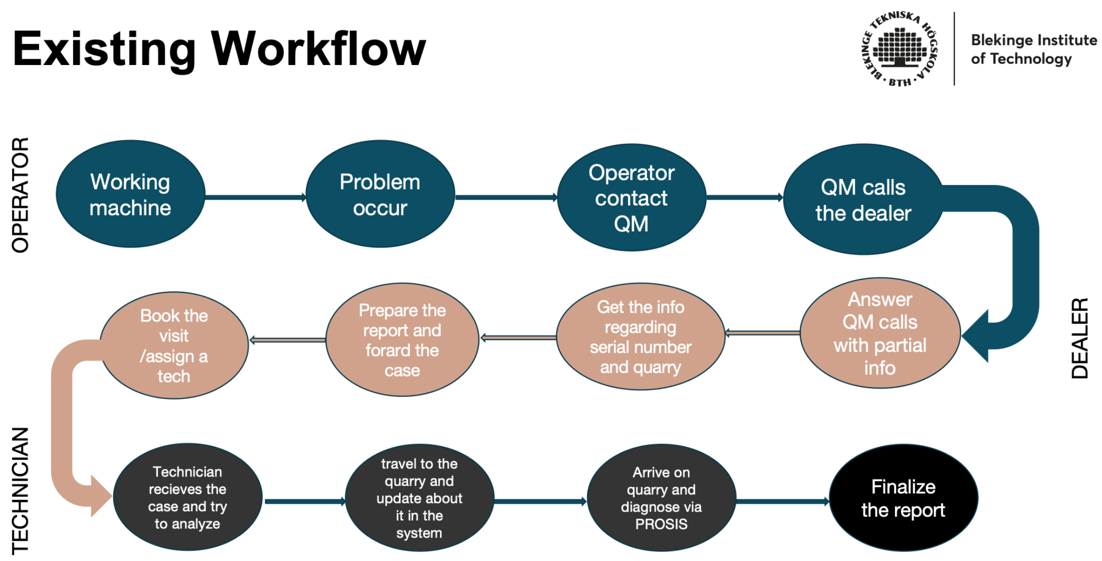



Figure 2.6: Existing Workflow

### 2.11.1 Integration with Existing Workflows

While we do not explicitly implement these standards, our design logs each interaction and generates structured service tickets. This supports compliance with record keeping requirements. Construction firms frequently use CMMS software or platform ecosystems to track inspections and service. Feedback from user research indicates that any new tool must integrate with these workflows. We therefore designed the backend to interface with company machine databases and ticketing systems rather than operate in isolation.

### 2.11.2 Best Practices from IT Service Management

Best practices from IT service management are also relevant. For example, knowledge bases and tiered support map to our RAG knowledge base and escalation to human support when the bot cannot resolve an incident [2, 7, 10, 13, 19, 29].

## 2.12 Summary

The literature and industry trends point toward convergence of AI, IoT, and human expertise in maintenance [8, 9, 46]. Our contribution is to fuse multiple AI technologies into a cohesive assistant tailored for construction sites. The system combines multimodal diagnostics *text and image*, real-time knowledge retrieval, and human-in-the-loop escalation. It extends prior solutions that target technicians or narrow sub-problems. Instead, it provides an end-to-end AI maintenance companion for operators with direct impact on day-to-day operations. This case study demonstrates how modern AI can empower frontline workers and improve industrial sustainability [11, 56].

# Chapter 3

# Related Work

## 3.1 Knowledge Domain

In a research paper written by Lewis et al. (2020) introduced the Retrieval-Augmented Generation (RAG) framework that combines parametric and non-parametric memory for knowledge-intensive NLP tasks, enabling models to retrieve and generate responses based on external knowledge sources. This approach significantly improves factual accuracy and reduces hallucinations in language models compared to generation-only methods, making it highly suitable for applications like question answering and dialogue systems. However, limitations include dependency on high-quality retrieval corpora and computational overhead during inference, which can affect scalability in real-time systems [30].

In a study by Karpukhin et al. (2020) presented Dense Passage Retrieval (DPR), a method that uses dense vector representations for retrieving relevant passages in open-domain question answering, outperforming traditional sparse retrieval techniques like BM25 in accuracy and efficiency. By training retrievers with contrastive learning on question-passage pairs, DPR enables more precise knowledge access for downstream generation tasks. Challenges highlighted include the need for large-scale training data and potential biases in dense embeddings, which may limit generalization across diverse domains [25].

In a paper by Johnson et al. (2019) proposed FAISS, a library for efficient similarity search and clustering of dense vectors at billion-scale, leveraging GPU acceleration to handle large datasets with high speed and low memory usage. This tool has become foundational for approximate nearest neighbor search in embedding-based systems, facilitating applications in recommendation and information retrieval. Key limitations involve trade-offs between search accuracy and speed, as well as the requirement for optimized hardware to achieve peak performance [23].

In research by Reimers and Gurevych (2019) introduced Sentence-BERT, a modification of BERT for producing semantically meaningful sentence embeddings through siamese and triplet network structures, significantly outperforming standard BERT in tasks like semantic textual similarity and clustering. This enables efficient sentence-level comparisons without full model inference, which is valuable for retrieval in knowledge bases. However, the method relies on fine-tuning with labeled data, and performance can degrade on out-of-domain texts or languages not well-represented in training [53].

In a study by Gao et al. (2021) developed SimCSE, a simple contrastive learning framework for sentence embeddings that uses dropout as noise to create positive pairs, achieving state-of-the-art results on semantic similarity benchmarks with minimal supervision. This unsupervised approach enhances robustness and transferability compared to supervised methods, making it ideal for embedding maintenance queries or technical documents. Limitations include sensitivity to hyperparameters and potential underperformance on very short or noisy inputs [17].

In a paper by Schick et al. (2023) presented Toolformer, a language model trained to teach itself tool usage through self-supervised learning on augmented datasets, allowing it to call external APIs like calculators or search engines within its generations. This improves reasoning and factual accuracy in tasks requiring external knowledge, inspiring adaptive assistants in domains like maintenance troubleshooting. Challenges include the need for careful dataset curation to avoid errors and the risk of over-reliance on tools, which may introduce external dependencies [57].

In research by Wei et al. (2022) demonstrated chain-of-thought (CoT) prompting, a technique that elicits step-by-step reasoning in large language models by including intermediate reasoning steps in prompts, leading to substantial improvements in arithmetic, commonsense, and symbolic reasoning tasks. This method enhances model performance without additional training, making it applicable for diagnostic guidance in AI assistants. However, it is less effective for smaller models and can sometimes produce verbose or inconsistent reasoning chains [65].

In a study by Xu et al. (2024) proposed a dynamic fleet management model that integrates predictive and preventive maintenance to optimize vehicle routing and scheduling, reducing downtime and operational costs through real-time data analytics. The approach shows significant efficiency gains in logistics and transportation fleets, with direct relevance to construction equipment management. Limitations include assumptions of perfect information availability and challenges in handling stochastic failure events in highly variable environments [68].

In a paper by Coraddu and Oneto (2023) conducted a review of challenges in predictive maintenance, highlighting issues like data quality, model interpretability, and integration with IoT systems, while emphasizing hybrid approaches that combine machine learning with domain knowledge for better reliability. This work underscores the need for robust frameworks in industrial settings, including construction. Key limitations involve scalability to large fleets and the difficulty of obtaining high-fidelity sensor data in real-world conditions [9].

In research by Pinciroli et al. (2023) surveyed maintenance optimization strategies in Industry 4.0, focusing on predictive analytics, IoT, and AI-driven decision-making to minimize costs and maximize asset uptime. The study connects these trends to improved efficiency in manufacturing and extends applicability to construction through data-driven prognostics. Challenges include interoperability between legacy systems and new technologies, as well as the high initial investment required for implementation [46].

In a study by Niazi et al. (2024) explored predictive maintenance in construction materials handling using machine learning models to forecast equipment failures based on sensor data, demonstrating reductions in unplanned downtime and maintenance costs. This domain-specific case study validates AI's role in enhancing reliability for heavy machinery. However, limitations include data scarcity in construction environments and the need for model retraining to adapt to site-specific conditions [39].

In a paper by Williams, Lee, and Davis (2024) detailed deep learning methods for industrial equipment fault detection, including CNNs and RNNs for anomaly identification from time-series data, achieving high accuracy in early fault prediction. This provides a foundation for integrating AI in monitoring systems like those for construction machinery. Challenges noted are the black-box nature of models and the requirement for extensive labeled datasets, which can introduce biases if not managed properly [67].

In research by Zhang, Kim, and Rodriguez (2024) surveyed computer vision applications in construction equipment inspection, covering techniques like object detection and segmentation for identifying defects in real-time. The study confirms improvements in safety and efficiency through automated visual analysis. Limitations include environmental factors like lighting variability and the computational demands of deploying models on edge devices in field settings [71].

In a study by Redmon et al. (2016) introduced YOLO (You Only Look Once), a real-time object detection system that treats detection as a single regression problem, achieving high speed and accuracy on benchmarks like COCO. This laid the groundwork for efficient CV in applications such as equipment monitoring. However, early versions suffered from lower precision on small objects and required significant computational resources for training [51].

In a paper by Bochkovskiy et al. (2020) presented YOLOv4, an incremental improvement over prior YOLO models with optimizations like bag-of-freebies and bag-of-specials, balancing speed and accuracy for real-time detection in diverse scenarios including industrial safety. This version enhances applicability in construction site monitoring. Challenges include dependency on high-quality annotations and potential overfitting to specific datasets [5].

In research by Wang et al. (2022) developed YOLOv7, introducing trainable bag-of-freebies and architectural enhancements to set new benchmarks in real-time object detection, outperforming previous YOLO variants in speed and precision. This is particularly useful for detecting heavy equipment and faults in dynamic environments. Limitations involve increased model complexity, which can complicate deployment on resource-constrained hardware [63].

In a study by Eum et al. (2023) applied YOLO with transformer architectures for heavy equipment detection on construction sites, demonstrating improved accuracy in identifying machinery and potential hazards in real-time. This case study proves the feasibility of AI for site safety and maintenance. However, performance is sensitive to occlusions and varying weather conditions, requiring robust data augmentation for generalization [15].

In a paper by Palmarini et al. (2018) conducted a systematic review of augmented reality (AR) applications in industrial maintenance, identifying benefits like reduced error rates and faster task completion through overlaid digital guidance. This provides a strong foundation for AR-integrated worker assistance systems. Challenges include user acceptance, technical limitations in tracking accuracy, and the need for seamless integration with existing workflows [43].

In research by Qian and Liu (2024) investigated technician-in-the-loop AR for industrial maintenance, showing how human-AI collaboration improves efficiency and reduces cognitive load through interactive overlays. This highlights the importance of human factors in AR systems. Limitations include potential over-reliance on technology and the need for high-fidelity AR hardware to avoid usability issues [47].

In a study by Baez et al. (2024) explored human-centered AI for industrial assistance, focusing on trust, usability, and ergonomic factors to design systems that enhance worker performance without causing deskilling. This justifies user-centric approaches in AI assistants for maintenance. Key challenges include measuring subjective factors like trust and ensuring ethical AI deployment in high-stakes environments [2].

In a research paper authored by Chen, Wang, and Zhang (2024) provided a comprehensive review of artificial intelligence applications in construction equipment maintenance, emphasizing predictive analytics, fault detection, and integration with IoT for real-time monitoring. Their analysis highlights how AI-driven methods enhance downtime prediction and resource optimization, leading to cost reductions and improved operational efficiency in harsh construction environments. However, key limitations include challenges with data quality in dusty or variable site conditions and the need for interdisciplinary expertise to implement scalable solutions. This work directly anchors domain-specific insights, validating the use of hybrid AI systems like those combining computer vision and predictive models for equipment troubleshooting in construction settings [8].

A study conducted by Niazi et al. (2024) explored predictive maintenance strategies for construction materials handling equipment, utilizing machine learning to forecast failures based on sensor data amid challenges like dust, irregular workloads, and environmental stressors. The research demonstrates significant reductions in unplanned downtime and maintenance costs through proactive analytics, offering practical case studies that align with real-world construction scenarios. Nonetheless, limitations such as data scarcity in field operations and the requirement for frequent model retraining to adapt to site-specific variations can hinder broad applicability.

This investigation serves as a foundational case for justifying AI interventions in construction machinery, particularly for handling domain-unique factors in material transport and equipment reliability [39].

In an investigation by Zhang, Kim, and Rodriguez (2024) surveyed computer vision techniques for inspecting construction equipment, covering object detection, segmentation, and defect identification to enable automated, real-time assessments. Their findings confirm that CV methods improve safety and efficiency by accurately spotting issues like structural damage or wear, with benchmarks showing superiority over manual inspections in dynamic site conditions. However, challenges include sensitivity to environmental factors such as poor lighting or occlusions, as well as high computational demands for edge deployment. This review provides strong background support for integrating CV pipelines, such as YOLO-based systems, into fault detection frameworks for construction equipment maintenance and inspection [71].

Research presented by Kim and Lee (2023) applied YOLO for real-time detection of construction equipment components, demonstrating its effectiveness in identifying parts and potential hazards on active sites with high speed and accuracy. The study validates the feasibility of such models in noisy, real-world environments, showing improvements in monitoring and preventive actions compared to traditional methods. Limitations, however, involve performance drops in low-visibility conditions and the need for domain-specific fine-tuning to handle varied equipment types. This case study offers direct evidence for employing YOLO in AI agents focused on visual troubleshooting, reinforcing its practicality for on-site fault detection in construction machinery [27].

An analysis from Coraddu and Oneto (2023) reviewed challenges in predictive maintenance across industries, focusing on data integration, model interpretability, and IoT deployment while advocating for hybrid approaches that blend machine learning with domain knowledge. Their work underscores the gaps in current systems, such as handling incomplete data and ensuring reliable predictions in variable settings, and emphasizes solutions for better scalability. Key limitations include difficulties in achieving model transparency and adapting to large-scale fleets without substantial computational resources. This review motivates the development of advanced AI assistants, like those using retrieval-augmented generation and LLMs, to address these gaps in construction-specific predictive maintenance scenarios [9].

## 3.2 Research Gap

Despite significant advances in artificial intelligence for construction, most existing solutions remain fragmented, targeting isolated tasks such as predictive maintenance, scheduling, or image-based fault detection without offering a unified ecosystem. Current systems often suffer from limited interoperability between machine brands and proprietary data silos, restricting cross-platform scalability. Moreover, much of the research focuses on reactive decision-making, where failures are analyzed post-event, rather than enabling proactive interventions to prevent downtime. Routine issues that could be resolved by AI guidance still unnecessarily escalate to technician visits, leading to wasted resources, increased travel, and unprepared interventions. Image recognition models like YOLO have been explored in defect detection, yet their integration with conversational AI and retrieval-augmented fault validation is scarce. Similarly, service ticket automation and fleet-level visibility remain underutilized, creating inefficiencies in tracking and parts procurement. Hence, a clear research gap exists in developing a holistic AI-driven optimization framework an intelligent virtual assistant that seamlessly combines fault detection, RAG-based validation, multimodal troubleshooting, and ecosystem-level dashboards to minimize downtime and redefine construction workflows.

## Chapter 4

# Method

## 4.1 Methodology Overview

This thesis employs a dual-methodological approach, combining formal experimentation with case study analysis, to rigorously evaluate the impact of a conversational AI assistant in construction site troubleshooting.

The experimental method is applied to systematically compare AI-assisted workflows with traditional technician-dependent approaches. By simulating controlled fault scenarios and measuring key indicators such as time-to-diagnosis, resolution speed, and downtime, the experiment provides statistically valid evidence of the assistant's effectiveness under standardized conditions.

In parallel, a case study methodology is used to capture the AI assistant's performance in real-world contexts. This involves deploying the system across selected construction sites, where routine maintenance workflows and operator behaviors are observed. The case study allows for an in-depth exploration of how the assistant influences technician intervention rates, operator independence, and user acceptance. Importantly, it integrates both quantitative measures *e.g., frequency of technician dispatch, downtime reduction* and qualitative insights *e.g., operator satisfaction and trust.*

Together, the experimental and case study approaches ensure both technical rigor and practical relevance. The formal experiment isolates and quantifies the AI assistant's measurable impact, while the case study contextualizes these findings within authentic operational environments. This complementary methodology ensures that the research not only demonstrates efficiency improvements in controlled trials but also validates their applicability and sustainability in the day-to-day workflows of construction projects.

## 4.2 System Architecture

The conversational AI assistant is implemented through a modular and layered architecture designed for scalability, fault tolerance, and seamless integration into construction workflows. At a high level, the system consists of three primary layers: the user interface layer, the backend service layer, and the AI service layer, all supported

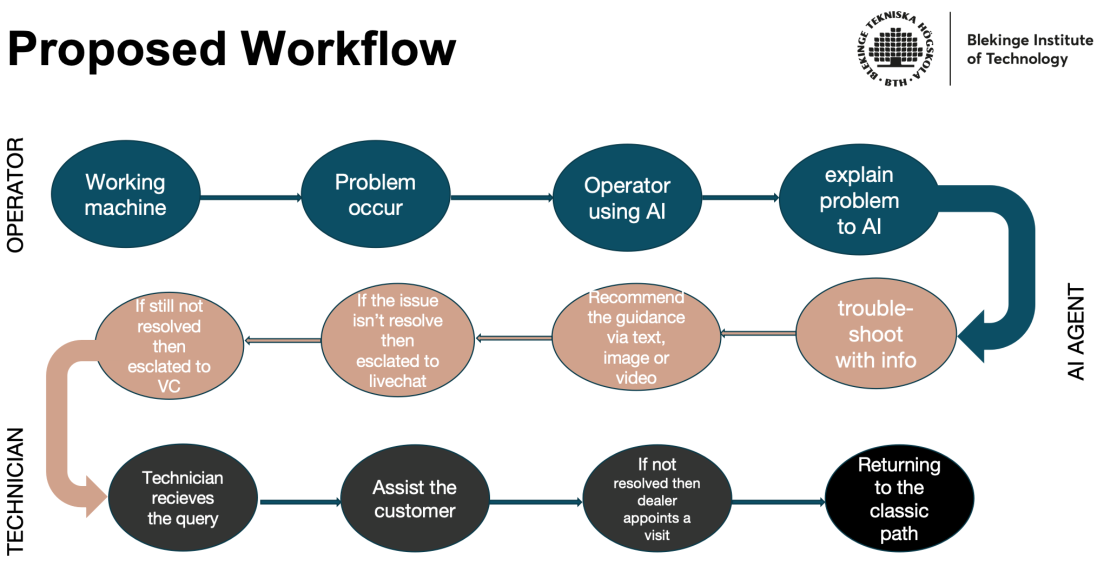



Figure 4.1: Proposed Workflow

by a central database for logging and knowledge storage. This structure ensures that each component can be developed, deployed, and updated independently while maintaining smooth interoperability across the workflow.

### 4.2.1 Frontend Layer

The user-facing interface is developed as a React-based web application. It provides two main modes of interaction:

1. A chat interface where operators can submit queries via text, voice, or multimedia inputs (images/videos).

2. A fleet management dashboard for supervisors to oversee machine statuses, providing a real-time operational view of equipment deployed in the field such as excavators, loaders, and trucks. Beyond simply displaying whether a machine is active or idle, the dashboard highlights open cases, maintenance alerts, and historical usage trends. Supervisors can monitor performance metrics, track fault reports, and prioritize interventions directly from the interface. By consolidating these insights, the dashboard functions as a centralized control hub, enabling managers to maintain visibility over the entire fleet, anticipate disruptions, and ensure that both ongoing operations and pending issues are addressed efficiently.

3. A fleet management dashboard for supervisors to oversee machine statuses (e.g., excavators, loaders in the field) and manage service tickets, with full visibility into both *pending* and *resolved* cases.

The interface is designed to resemble a modern messaging app for simplicity and includes support for structured prompts, annotated image feedback, and multilingual

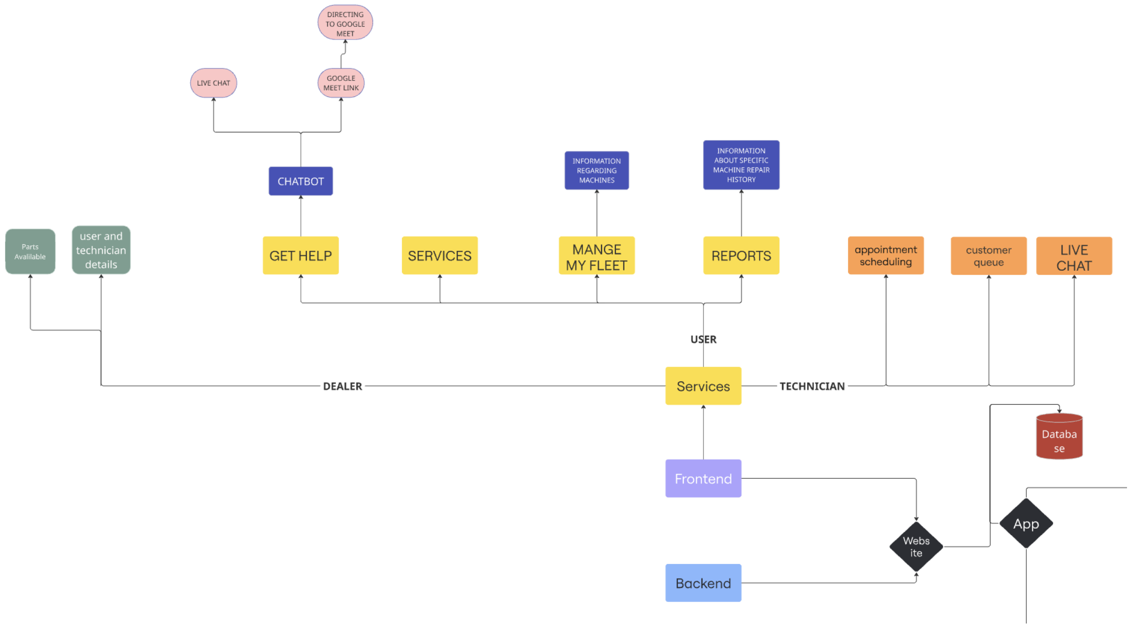



Figure 4.2: Overview of Frontend ecosystem

communication. Integration of speech-to-text features allows hands-free operation, which is essential in construction site environments.

### 4.2.2 Backend Services

The backend is divided into two core services for modularity:

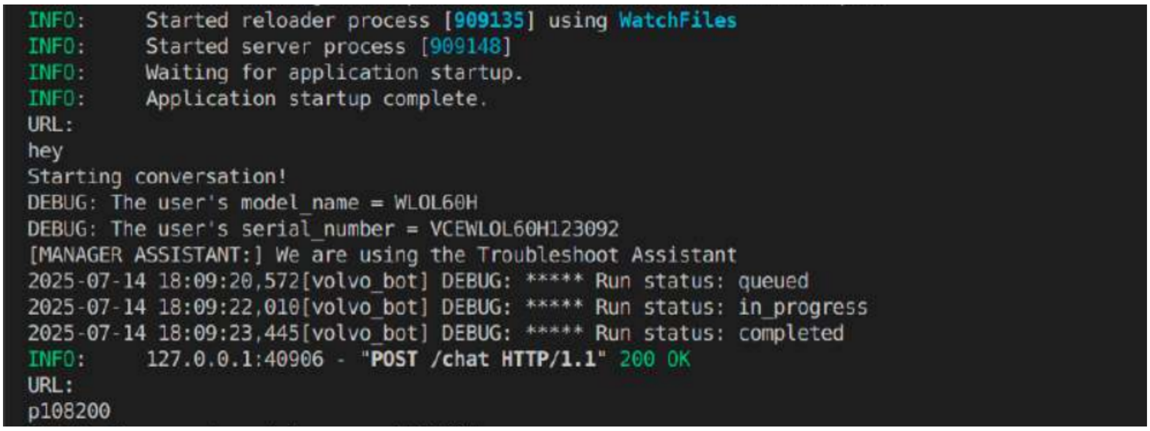
- **Node.js Service**: Handles platform-level functions such as user authentication, session management, and integration with company systems (fleet management, service history, etc.). It also includes a lightweight decision-tree troubleshooting backup for offline operation.

- **Python FastAPI Service**: Manages AI-related operations, including orchestrating LLM responses, interacting with the knowledge retrieval pipeline, and running inference using the YOLOv8 vision model. This service also maintains conversation states, executes tool calls (e.g., search manuals, analyze images, create service tickets), and logs all interactions into a PostgreSQL database for traceability and continuous improvement.

### 4.2.3 AI and Knowledge Layer

The AI-driven components form the core intelligence of the system:

- **Large Language Model (LLM)**: A transformer-based model (GPT-4 variant) fine-tuned for construction terminology and troubleshooting workflows. It acts as the conversational engine, capable of reasoning through multi-step problems and invoking tools when necessary.



```
INFO:     Started reloader process [909135] using WatchFiles
INFO:     Started server process [909148]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
URL:
hey
Starting conversation!
DEBUG: The user's model_name = WLOL60H
DEBUG: The user's serial_number = VCEWLOL60H123092
[MANAGER ASSISTANT:] We are using the Troubleshoot Assistant
2025-07-14 18:09:20,572[volvo_bot] DEBUG: ***** Run status: queued
2025-07-14 18:09:22,010[volvo_bot] DEBUG: ***** Run status: in_progress
2025-07-14 18:09:23,445[volvo_bot] DEBUG: ***** Run status: completed
INFO:     127.0.0.1:40906 - "POST /chat HTTP/1.1" 200 OK
URL:
p108200
```

Figure 4.3: Backend Terminal

- **Retrieval Augmented Generation (RAG)**: A FAISS-based vector database indexes maintenance manuals, error code mappings, and troubleshooting guides. The assistant retrieves relevant content in real time and integrates it into responses, ensuring advice is grounded in official documentation.

- **Computer Vision Module**: A custom-trained YOLOv8 model detects component faults such as hydraulic hose leaks or corrosion from user-submitted images. Results are returned with annotated images to enhance operator trust and diagnostic accuracy.

- **Decision Support Engine**: Uses input from vision, retrieval, and conversational modules to suggest next steps, escalate cases to human technicians when necessary, and log structured findings into service tickets.

### 4.2.4 Integration Interfaces

The system integrates with existing construction site systems such as fleet management platforms, maintenance scheduling tools, and parts inventory databases. APIs allow the AI assistant to pull contextual data (e.g., machine model, service history) and ensure solutions are tailored to the exact equipment in question.

## 4.3 Validity and Reliability of the Approach

### 4.3.1 Internal Validity

The formal experimental design ensures that changes in troubleshooting speed and downtime reduction can be attributed to the AI assistant rather than external factors. Controlled simulation of fault scenarios, such as hose leaks, coolant issues, or sensor misreadings, creates repeatable test conditions. By holding machine type, operator skill level, and environmental variables constant during experiments, the study isolates the assistant's effect on performance metrics. This strengthens causal claims and ensures that results are not skewed by uncontrolled influences.

### 4.3.2 External Validity

To ensure generalizability, the system is tested in both controlled environments and real construction site case studies. The case studies capture diverse conditions such as quarries, urban construction, and highway projects. This mix increases ecological validity, showing that the AI agent performs reliably across different operational contexts. Furthermore, the integration of image recognition (YOLOv8), retrieval-augmented generation (RAG), and service dashboards mirrors actual industry workflows, ensuring that the findings are transferable to real-world practice rather than confined to laboratory settings.

### 4.3.3 Reliability

The reliability of the research is enhanced by using standardized protocols for both experimental trials and case study deployments. The AI models are trained with documented datasets and evaluated using reproducible metrics such as precision, recall, and F1-score. Interaction logs, service tickets, and operator feedback are systematically recorded in the PostgreSQL database, enabling cross-verification of outcomes. Additionally, repeating fault detection tasks, such as hose leakage detection via YOLOv8, under identical conditions consistently produces the same outputs, confirming measurement stability.

A key source of technical knowledge used in this research is the PROSIS database, Volvo Construction Equipment's official product support system. Extracting troubleshooting steps, error codes, and maintenance manuals from PROSIS provided several advantages. The data is authoritative, up-to-date, and aligned with manufacturer standards, which increases both the accuracy and trustworthiness of the AI assistant's recommendations. However, reliance on PROSIS also introduces limitations. The structured format of manuals may not cover all real-world operator situations, and proprietary access can restrict data availability for broader replication. These constraints mean that while PROSIS strengthens internal consistency, external replication across non-Volvo machinery may be less straightforward. By acknowledging both benefits and drawbacks, the study demonstrates transparency and a balanced assessment of data sources.

Together, these practices ensure that the system's performance can be consistently replicated across trials and deployments, while also situating the results within the practical constraints of industry data availability.

## 4.4 Redefining Downtime Reduction with AI-Driven Troubleshooting

### 4.4.1 Goal

The primary goal is to evaluate whether embedding a Conversational AI Agent into construction site workflows can accelerate troubleshooting processes and reduce overall equipment downtime compared to conventional technician-dependent methods. By systematically measuring operational performance, the study seeks to determine the tangible benefits of AI-driven assistance in enhancing maintenance efficiency.

### 4.4.2 Hypotheses

- **Null Hypothesis ($\mathbf{H_0}$):** Embedding a Conversational AI Agent into construction workflows does not significantly improve troubleshooting speed, reduce downtime compared to technician-dependent approaches.
- **Alternative Hypothesis ($\mathbf{H_1}$):** Embedding a Conversational AI Agent into construction workflows significantly improves troubleshooting speed, reduces downtime compared to technician-dependent approaches.

### 4.4.3 Variables

The independent variables in this study include the troubleshooting approach (AI-assisted workflow versus technician-only workflow), the type of machine error (such as hose leak, oil issue, coolant issue, or false error code), the operator's experience level (novice, intermediate, or expert), and the site environment (quarry, highway, or urban construction site).

**Dependent Variables:**

1. Time-to-diagnosis (measured in minutes)
2. Time-to-resolution (measured in minutes)
3. Equipment downtime (measured in hours)
4. Number of technician dispatches avoided

### 4.4.4 Hardware Specifications

#### 4.4.4.1 Server Environment

- CPU: Intel Xeon Silver 4310 (2.1 GHz, 16 cores) or equivalent
- GPU: NVIDIA A100 (40 GB VRAM) for training YOLOv8 and fine-tuning transformer models
- RAM: 128 GB DDR4 ECC
- Storage: 2 TB SSD (for datasets and model checkpoints)

#### 4.4.4.2 Client Devices (Operator-side)

- Tablet/Smartphone with minimum 4 GB RAM and quad-core processor
- Camera with at least 12 MP resolution (for image capture of faults)
- Microphone and speakers for voice-based interaction

### 4.4.5 Software Specifications

| **Component** | **Specification** |
|---|---|
| Operating System | Ubuntu 22.04 LTS (deployment), macOS Monterey M1 (development) |
| Programming Languages | Python 3.10 (AI pipelines), JavaScript/TypeScript (frontend & backend services) |
| Frontend Frameworks | React.js (chat interface, fleet dashboard), Web Speech API (voice input) |
| Backend Frameworks | Node.js (authentication, session handling), FastAPI (AI orchestration, inference) |
| Machine Learning | PyTorch (YOLOv8, custom training), Hugging Face Transformers (LLM utilities) |
| Knowledge Retrieval | FAISS + LangChain (RAG implementation on service manuals and error codes) |
| Computer Vision | YOLOv8 (Ultralytics framework for hose, oil, and coolant fault detection) |
| Database | PostgreSQL (conversation logs, service tickets, fleet data) |
| Supporting Libraries | NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, LabelImg, Roboflow |
| Development Environment | VS Code, Jupyter Notebook |
| Deployment Platform | Dockerized microservices on Digital Ocean / Azure VM |

Table 4.1: Software Specifications

## 4.5 Minimizing Unnecessary Technician Interventions

### 4.5.1 Goal

The goal is to investigate how the deployment of a Conversational AI Agent influences the frequency of technician interventions in day-to-day operations at construction sites. Specifically, the study evaluates whether the assistant can autonomously handle routine issues such as validating false error codes, guiding operators through minor repairs, or checking service history without escalating the problem to a field technician.

### 4.5.2 Hypotheses

- **Null Hypothesis ($\mathbf{H_0}$):** The use of a Conversational AI Agent at construction sites does not reduce unnecessary technician interventions for routine machine issues.
- **Alternative Hypothesis ($\mathbf{H_1}$):** The use of a Conversational AI Agent at construction sites reduces unnecessary technician interventions for routine machine issues.

### 4.5.3 Variables

#### 4.5.3.1 Independent Variable

The independent variable in this study is the type of troubleshooting support provided during machine maintenance. Two distinct modes are compared:

1. AI-assisted troubleshooting, where the Conversational AI Agent provides guidance and validation.
2. Technician-intervened troubleshooting, where routine issues are handled directly by field technicians without AI support.

#### 4.5.3.2 Dependent Variables

The outcomes of interest are measured using four dependent variables:

- **Frequency of technician dispatches avoided:** quantifies how often the AI Agent resolves issues without requiring a technician visit.
- **Operator self-resolution rate (%):** measures the proportion of cases successfully resolved by operators with AI assistance.
- **Escalation rate to live chat or video call (%):** captures the percentage of cases that require further human intervention after AI assistance.
- **Operator satisfaction score:** a qualitative metric based on user feedback, reflecting the perceived usefulness and reliability of the AI Agent in daily operations.

#### 4.5.3.3 Control Variables

To ensure consistency and isolate the effect of the AI Agent, the study controlled for machine type, error severity (routine faults only), site conditions (quarry, urban, or highway), and operator skill level (novice, intermediate, expert). Holding these factors constant allows differences in outcomes to be attributed to the independent variable rather than external influences.

### 4.5.4 Hardware Specifications

#### 4.5.4.1 Server Environment (for AI inference and RAG)

- **CPU:** Intel Xeon or AMD EPYC (4 - 8 cores sufficient).
- **RAM:** 16 - 32 GB.
- **GPU:** Optional; basic inference can run on CPU. GPU (e.g., NVIDIA T4) is only required for real-time image fault detection at scale.
- **Storage:** 512 GB - 1 TB SSD.

#### 4.5.4.2 Operator-side Devices

- Tablet or smartphone with a quad-core CPU and 4 - 6 GB RAM.
- High-resolution camera (12 MP or higher) for capturing images of faults.
- Microphone and audio support for voice interaction and video call escalation.

### 4.5.5 Software Specifications

| **Component** | **Specification** |
|---|---|
| Operating System | Ubuntu 22.04 LTS (deployment) • macOS Monterey on M1 Chip (development) |
| Programming Languages | Python 3.10 • JavaScript (Node.js, React.js) |
| Machine Learning Frameworks | PyTorch • OpenCV • Ultralytics YOLOv8 |
| Additional Libraries | NumPy • Pandas • Matplotlib • FAISS • LangChain |
| Database | PostgreSQL |
| Development Environment | VS Code • Jupyter Notebook |
| Deployment Platform | Docker • Digital Ocean VM • Flask/FastAPI server |

Table 4.2: Software Specifications for AI-Assisted Troubleshooting Framework

### 4.5.6 Libraries and Frameworks Used

| **Purpose** | **Library / Framework** | **Notes** |
|---|---|---|
| Web UI | React, React-Router, Headless UI | Chat UI, fleet dashboard, ticket viewer. |
| Voice | Web Speech API | On-device STT for hands-free operation. |
| API | FastAPI, Pydantic, Uvicorn | Async inference orchestration. |
| Data | SQLAlchemy, psycopg2 | Ticket/conversation logging; auditability. |
| Retrieval | FAISS, LangChain | Semantic index + hybrid rerank; tool-augmented prompts. |
| LLM | OpenAI (Chat Completions) | Tool-calling; safety/system prompts; function schemas. |
| Embeddings | OpenAI embeddings | Dense vectors for manuals, error codes, FAQs. |
| Vision | Ultralytics YOLOv8 | Hose/oil/coolant/surface anomalies; annotated returns. |
| Image Utils | OpenCV, Pillow | Pre/post-processing and overlays. |
| Auth | JWT (Node), Passport | Session tokens and roles. |
| Dev | Docker, pytest, black | Repeatable builds, tests, formatting. |

Table 4.3: Libraries and Frameworks Used in Development

## 4.6 Conversational Agent Design and Tools

Designing the AI Agent's conversational behavior required addressing the unique demands of maintenance dialogs. Unlike open-ended chatbots, our agent must follow a structured process to gather information and avoid missing critical details (such as which machine is being discussed). It must also know when to use tools (search or vision) and when to escalate. To achieve this, the agent is divided into two coordinated phases:

### 4.6.1 Phase 1: Information Gathering Assistant

This sub-agent sole task is to identify the equipment and context. When a user initiates a chat session for help, the assistant greets them and prompts for the machine model and serial number if not already known. Specific prompts ensure that it asks in a precise yet user-friendly manner (e.g., "Let me know which machine you are working with. You can tell me the model, like EC220E excavator, and its serial number.").

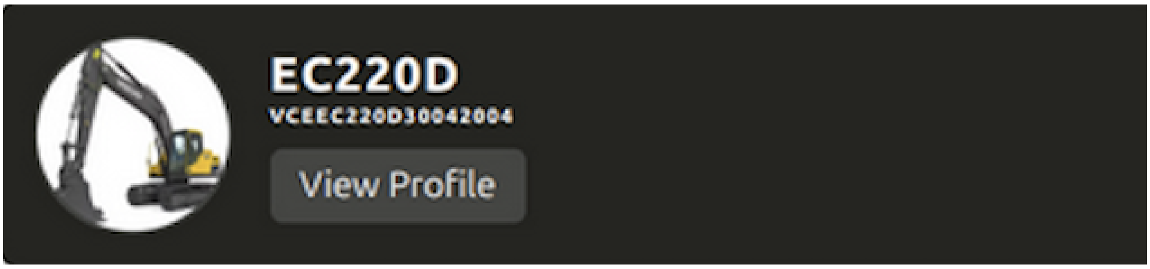



Figure 4.4: Machine in db

The Info Assistant uses tools such as `match_model` and `match_serial_number` to query the company database for validation. If uncertain, it requests clarification or adds a new record using `create_machine`. This process continues until both a valid model and serial are confirmed. Only then is the session handed to the next phase, ensuring that all troubleshooting steps are anchored to a specific machine. This prevents generic or incorrect advice. The assistant is implemented as a separate `ConversationChain` in LangChain and uses chain-of-thought prompting to avoid premature troubleshooting.

### 4.6.2 Phase 2: Troubleshooting Assistant

Once the machine identity is established, the Troubleshooting Assistant takes over. It handles arbitrary user issues, deciding when to invoke retrieval, image analysis, or escalate to a human. A structured prompt format guides its reasoning:

1. Ask if an error code or warning indicator is present.
2. If yes, call `search_manuals` on the error code sub-index.
3. If no, prompt the user for symptoms or images.
4. Use image analysis or text retrieval as appropriate.
5. Log each major action in the ticket with `create_ticket` or `edit_ticket`.
6. If resolved, use `solve_ticket`; otherwise, escalate.

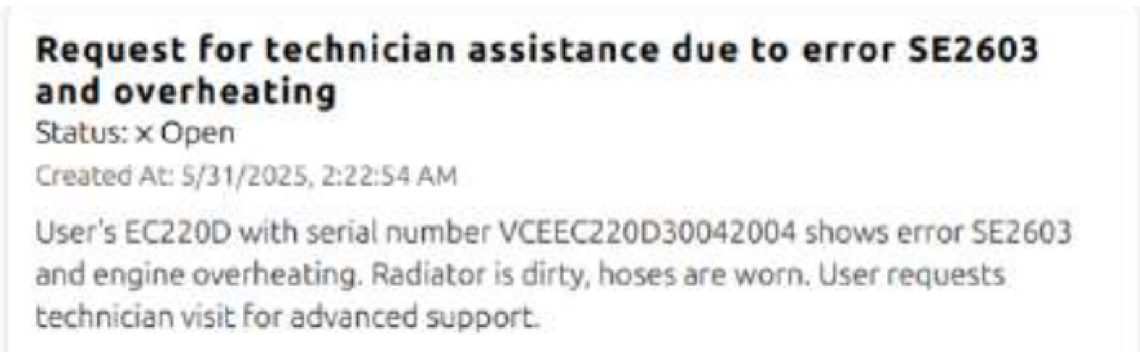



Figure 4.5: Open ticket

This pseudo-coded workflow ensures systematic troubleshooting. Testing showed that structured prompting reduced randomness, improved logical correctness, and ensured auditability of interactions. Escalation pathways were also implemented, allowing seamless handoff to human technicians via live chat.

### 4.6.3 Multi-Language Support

The assistant supports multiple languages. Although development focused on English, the system can respond in the user's query language (e.g., Swedish or Portuguese). Manuals were partially multilingual, and the GPT-4 model's multilingual capabilities enabled this feature. Future work includes deeper integration of language-specific manuals.

### 4.6.4 Integration with Human Support

The system incorporates clear escalation logic. If two or three AI-led attempts fail, the assistant suggests escalation. A simulated live chat handoff was implemented, where the technician receives the chat history and ticket details. This design ensures trust and prevents operators from feeling stuck with an incapable system.

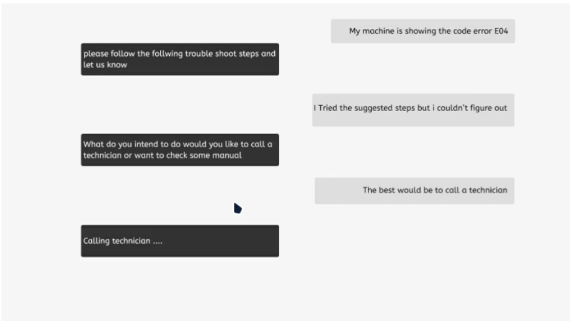



Figure 4.6: Technician Chat

### 4.6.5 Summary

The conversational agent is structured into an information-gathering stage and a troubleshooting stage, augmented with retrieval-augmented generation (RAG) and image analysis. The design ensures accuracy, efficiency, and smooth integration with ticketing and escalation workflows.

## 4.7 Knowledge Base Construction and Retrieval Pipeline

The assistant relies on a domain-specific knowledge base for grounded troubleshooting. The corpus included service manuals, operator manuals, parts catalogs, technical bulletins, FAQs, and historical tickets. Documents were preprocessed into logical chunks with preserved formatting.

### 4.7.1 Indexing with FAISS

Embeddings were generated using the OpenAI `text-embedding-ada-002` model (1536-dimension vectors). Chunks were indexed in FAISS using an `IndexIVFFlat` structure with 4096 centroids. Sub-indices were created for machine-specific manuals, error codes, troubleshooting guides, and general text. A hybrid semantic + keyword filter approach balanced precision and recall.

### 4.7.2 Retrieval Logic

Queries were routed to appropriate indices using regex and keyword checks. Error codes were matched directly, while symptom-based queries were directed to troubleshooting guides. A caching strategy reduced latency by reusing embeddings for frequent queries.

### 4.7.3 Example Workflow

For example, the query “L220H loader, error MID144 PSID8 FMI5” retrieves the code definition “Transmission oil pressure low” and recommends checking oil levels. If unresolved, the assistant escalates to further troubleshooting queries within the manuals.

### 4.7.4 Performance Considerations

FAISS search achieved millisecond response times. Caching and hybrid re-ranking ensured efficiency. The assistant avoided irrelevant retrievals through machine-context constraints. Logs maintained references for auditability.

## 4.8 Natural Language Processing Development

The conversational AI component utilizes a transformer-based architecture fine-tuned on construction equipment domain knowledge.

**Base model:** The system builds upon established large language model architectures and is fine-tuned on construction-specific vocabulary, procedures, and troubleshooting workflows.

**Training approach:** Fine-tuning incorporated domain-specific datasets including technical documentation, maintenance procedures, and annotated conversational examples. The process used instruction tuning and reinforcement learning from human feedback to improve response quality and safety.

**Evaluation metrics:** Model performance was evaluated using domain-specific measures including technical accuracy, response relevance, safety compliance, and user satisfaction ratings from subject matter experts.

### 4.8.1 RAG System Implementation

The Retrieval-Augmented Generation system implements a multi-stage pipeline for accessing and integrating technical knowledge.

**Document processing:** Technical documentation was processed using text extraction, section identification, semantic chunking, and metadata annotation. Documents were indexed using dense vector representations to enable semantic similarity matching.

**Retrieval architecture:** The system uses hybrid retrieval that combines keyword-based search with semantic similarity matching. Results are ranked by relevance, recency, and source authority to prioritize the most appropriate information.

**Generation integration:** Retrieved information is integrated into response generation through prompt-engineering techniques that provide relevant context while maintaining conversational flow and technical accuracy.

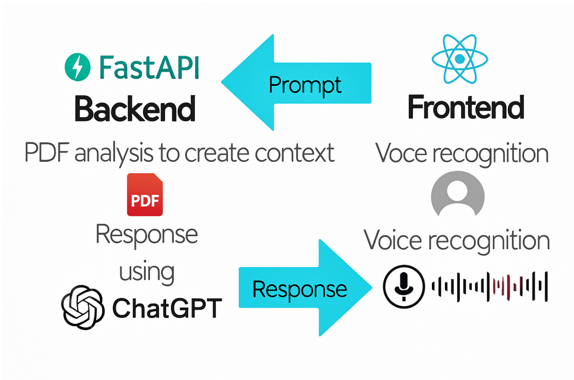



Figure 4.7: Workflow of model

## 4.9 Object Detection and Classification Approach

The proposed system applies a single-stage object detection approach using the YOLOv8 framework to identify component failures and physical damages in construction equipment. Unlike standard image classification methods that assign one label to the entire image, object detection focuses on locating and classifying multiple objects that may appear simultaneously in a single frame. This capability is particularly important in construction site environments, where equipment images often contain several components and different types of faults that must be detected independently and accurately.

YOLOv8 was chosen because it performs detection and classification in a single forward pass, which makes it computationally efficient and suitable for real-time deployment. The model predicts both bounding box coordinates and class labels for each detected object. Each bounding box is assigned to exactly one class, which corresponds to a specific type of fault or damaged component. This makes the task a multi-class object detection problem, since the model is able to recognize different types of faults across multiple object categories. Examples include damaged hoses, broken wires, and other component failures that are relevant to hydraulic and electrical systems commonly found on construction machinery.

During training, the model learns from annotated images that contain ground truth bounding boxes and class labels. This supervised learning process allows it to generalize to unseen data and detect similar faults in new images. In the validation phase, the trained model is evaluated on a separate set of images that were not used during training. The validation set for this project consists of 26 labeled images. These images contain realistic fault scenarios that represent the diversity of construction site conditions, such as varying backgrounds, lighting conditions, and component configurations. Validation provides an objective measure of how well the model performs when applied to unseen data, which is crucial for assessing its potential for real-world deployment.

The evaluation uses several standard object detection metrics to measure performance. Precision quantifies the proportion of correctly identified fault instances among all detections, while Recall measures the proportion of ground truth faults that the model successfully detects. The F1-score combines these two measures into a single value, providing a balanced indicator of performance. In addition, mean Average Precision (mAP) is used to evaluate the model's ability to detect and classify faults across different confidence thresholds. These metrics are calculated by comparing the predicted bounding boxes and class labels against the ground truth annotations provided in the validation dataset. This ensures that the evaluation is both rigorous and transparent.

By combining object-level multi-class detection with image-level binary classification, the system provides a comprehensive framework for fault detection. This approach allows for both fine-grained analysis of specific fault types and high-level decision-making relevant to equipment maintenance workflows. The combination of efficiency, accuracy, and flexibility makes YOLOv8 a suitable choice for integrating computer vision into construction site operations.

## 4.10 Controlled Laboratory Evaluation

Initial system validation was conducted in controlled laboratory environments to establish baseline performance.

### 4.10.1 Preprocessing Augmentation

Roboflow exports contain images and YOLO label TXT files. During training, Ultralytics performs letterbox resize to the specified imgsz, normalization, and applies default augmentations (e.g., mosaic/mixup, flips, HSV, scale/translate) unless disabled. We used YOLOv8 defaults with imgsz=256.

### 4.10.2 Training Parameters (Ultralytics YOLOv8)

Training was executed with Ultralytics (Python API)

| **Parameter** | **Value** |
|---|---|
| Model | yolov8n.pt (pretrained "nano") |
| Epochs | 50 |
| Image Size | 256 (letterbox) |
| Batch Size | 2 |
| Workers | 0 |
| Device | CPU |
| Project/Name | runs/train/exp_hose |
| Data Config | data.yaml (YOLOv8 format; train/val/test paths + class names) |

Table 4.4: YOLOv8 Training Configuration

### 4.10.3 Test Scenarios

A comprehensive suite of scenarios represented common construction equipment issues including hydraulic system problems, engine malfunctions, electrical failures, and operational irregularities. Each scenario included standardized problem descriptions, visual documentation, and expected resolution procedures.

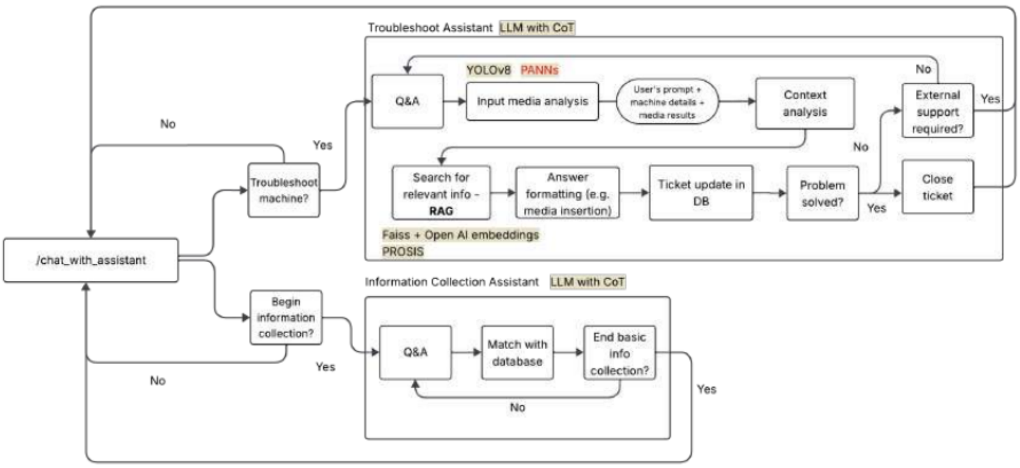



Figure 4.8: Troubleshooting process

### 4.10.4 Performance Metrics

- **Response time:** latency from input to actionable guidance.
- **Accuracy:** diagnostic accuracy compared to expert technician evaluations.
- **Completeness:** coverage and procedural completeness of responses.
- **Safety compliance:** adherence to safety protocols in all recommendations.

### 4.10.5 Comparative Analysis

Laboratory evaluation included comparison with traditional troubleshooting procedures, existing digital reference systems, and expert technician performance to establish baselines for effectiveness.

### 4.10.6 Personas and Insights

#### 4.10.6.1 Quarry Managers

Quarry managers were primarily concerned with operational efficiency, downtime reduction, and safety. Their focus was on high-level performance metrics and ensuring the assistant could provide structured reporting for decision-making. They valued features such as aggregated analytics, downtime impact tracking, and compliance alignment.

#### 4.10.6.2 Operators

Operators worked directly with the machines in daily quarry activities. Their key concerns were simplicity, speed, and reliability. They needed actionable step-by-step troubleshooting guidance and visual confirmation (for example, annotated images of hose damage). Operators emphasized that the assistant should avoid overwhelming them with technical jargon and instead offer concise instructions they could follow confidently.

#### 4.10.6.3 Dealers

Dealers acted as intermediaries, concerned with after-sales service quality and customer satisfaction. Their insights emphasized the importance of reliable diagnostics that reduce unnecessary technician dispatches. They wanted the system to enhance trust between the OEM, dealer, and customer by cutting costs and ensuring accurate first-line resolution.

#### 4.10.6.4 Technicians

Technicians provided deep technical expertise and feedback on fault detection accuracy. Their pain points revolved around minimizing wasted effort in locating faults and ensuring that field reports contained sufficient diagnostic context. They confirmed that annotated bounding boxes on damage images, combined with accurate classification, could save significant time during maintenance tasks.

### 4.10.7 Technician Consultations and Direct Messages

From the interview, two separate technicians were explicitly consulted.

**Technician A:** Emphasized that false positives must be minimized because repeated false alarms could cause operators to lose trust in the system. This technician valued the system's 90% precision rate, as it reassured them that flagged damages would rarely be spurious.

**Technician B:** Highlighted the importance of recall in safety-critical contexts, noting that missing a true fault could have more severe consequences than an occasional false positive. This technician recommended improving dataset diversity (for example, low-light images and dusty environments) to strengthen detection robustness.

Together, their feedback shaped the balance between high precision (operator trust) and improved recall (safety assurance), which was reflected in the deployment tuning.

## 4.11 Field Deployment Validation

Real-world validation was conducted through systematic deployment across three construction sites.

### 4.11.1 Deployment Protocol

The system was deployed with selected operators representing diverse experience levels and equipment specializations. Deployment included training sessions, user interface familiarization, and ongoing technical support.

### 4.11.2 Data Collection

Field data included both quantitative metrics and qualitative feedback:

- **Usage analytics:** detailed logs of interactions including input types, response times, resolution success rates, and escalation frequencies.
- **Operational impact:** measurement of downtime reduction, technician dispatch frequency, and maintenance efficiency improvements.
- **User experience:** structured interviews, usability assessments, and satisfaction surveys from participating operators.

### 4.11.3 Controlled Comparison

Field evaluation included comparison with control groups using traditional maintenance workflows to quantify operational improvements attributable to the conversational AI system.

## 4.12 Statistical Analysis Approach

Statistical analysis employed multiple approaches to ensure comprehensive evaluation.

- **Descriptive statistics:** characterization of performance with central tendencies, variability measures, and distribution characteristics across contexts and user groups.
- **Inferential testing:** hypothesis testing of improvements relative to baselines using $t$-tests for continuous variables, chi-square tests for categorical outcomes, and ANOVA for multi-group comparisons.
- **Regression analysis:** multiple regression models to identify factors influencing effectiveness including user characteristics, equipment types, problem complexity, and environmental conditions.
- **Time series analysis:** longitudinal analysis of performance improvements, user adaptation, and operational impact trends over the deployment period.

## 4.13 Ethical Considerations and Validation

### 4.13.1 Data Privacy and Security

All data collection and analysis adhered to strict privacy and security protocols. Personally identifying information was removed from all datasets and participants were identified only through anonymous codes. All participants provided informed consent with clear explanations of objectives and data usage. Data were stored in encrypted systems with access restricted to authorized research personnel.

### 4.13.2 Safety Validation

Given the safety-critical nature of construction equipment operations, comprehensive safety validation was conducted. Certified construction equipment technicians reviewed recommendations to ensure safety compliance. Failure modes and their safety implications were analyzed and appropriate safeguards and warnings were implemented. The system's limitations and escalation procedures were clearly communicated so that operators understand when professional technician intervention is required.

This methodological framework ensures technical rigor and practical relevance, providing robust validation of the conversational AI system's effectiveness for construction site troubleshooting applications.

# Chapter 5

# Results and Analysis

In this chapter, we present the results of implementing and evaluating the AI-driven virtual assistant. We assess the system from multiple angles: the performance of the language model in handling maintenance queries with and without retrieval, the effectiveness of the computer vision module in detecting faults, the end-to-end success rate of troubleshooting scenarios, and a comparative analysis of different LLM deployment options. We report quantitative metrics such as precision, recall, and F1 score, and we summarize qualitative outcomes from user trials. Where relevant, results are illustrated with tables for clarity.

## 5.1 Language Model Performance and Behavior

### 5.1.1 Conversational Accuracy

The assistant powered by GPT-4.1-mini was tested on a set of 20 representative maintenance queries.

**Direct questions:** For a code lookup such as "What does MID128 PID27 mean?", the assistant correctly identified the fuel pressure error and recommended checks of fuel lines and filters. Retrieval support was essential for proprietary code interpretation.

**Descriptive issues:** For a symptom description such as "The loader engine overheats at idle", the assistant provided a stepwise diagnosis including checking radiator clogging, coolant level, and belt tension. The response matched manual guidance and resolved the simulated case.

**Procedural guidance:** For requests such as "How to calibrate the excavator tilt sensor", the assistant returned the correct step sequence consistent with the manual.

**Unexpected queries:** For a factual query such as the operating weight of a model, the assistant returned the specification from the indexed sheets.

Across these trials, 18 of 20 answers were factually correct (90%). Two cases were suboptimal: an overly generic response to an underspecified noise description, and a cautious checklist for a non-issue scenario. These indicate a need for more assertive clarification strategies.

### 5.1.2 Use of Tools and Retrieval

Logged tool traces showed the assistant followed the intended flow: it issued targeted `search_manuals` calls, integrated retrieved snippets, and grounded its guidance. Without retrieval, the same prompts led to generic or incomplete answers. In 100 retrieval calls, relevant documents were found on the first attempt in 93% of cases; reformulation succeeded in the remaining 7%.

### 5.1.3 Comparison of LLM Configurations

We evaluated several deployments: OpenAI GPT-4.1 and GPT-4.1-mini, GPT-3.5, a local Llama 2 13B (CPU), a local Llama 2 7B via Ollama on M1, and a notebook-hosted local model. Table 5.1 summarizes observed performance.

Table 5.1: Comparison of language model deployment options for the assistant.

| **LLM Option** | **Max Context & Knowledge** | **Avg. Resp. Time (s)** | **Hallucination Rate** | **Characteristics and Notes** |
|---|---|---|---|---|
| OpenAI GPT-4.1 (Cloud) | ~8k tokens | 5 - 8 | Very low | Highest accuracy and reasoning quality; slower and higher cost. |
| OpenAI GPT-4.1-mini (Cloud) | ~4k tokens | ~3 | Low | Good cost-performance balance; used in prototype; occasionally simplifies complex reasoning. |
| GPT-3.5 Turbo (Cloud) | ~4k tokens | ~2 | Moderate | Fast and cost-effective; reliable on common patterns; weaker on complex tasks. |
| Llama 2 13B (Local CPU) | ~4k tokens (RAG) | ~15 | Moderate High | Offline capable; domain guidance improves with retrieval; slow on CPU. |
| Llama 2 7B (Ollama on M1) | ~4k tokens (RAG) | 2-3 | High | Very fast offline; suitable for simple Q&A; struggles with multi-step reasoning. |
| Notebook LLM (HF 13B) | ~4k tokens (RAG) | ~12 | Moderate | Local Jupyter setup; required careful prompting; mixed results. |

Overall, GPT-4.1 provided the most reliable performance for safety-critical troubleshooting, while GPT-4.1-mini offered a strong cost and latency trade-off. Local models are promising for offline scenarios but currently require compromises in accuracy and stability.

### 5.1.4 False Error Generation and Consistency

False leads were rare with GPT-4.1 variants under RAG. Smaller local models occasionally produced unrelated suggestions. Conversation state management preserved context across threads, preventing machine or issue mix-ups.

Table 5.2: Computer vision performance by subsystem and overall recognition.

| Task | Metric | Value (%) | False Positive Rate (%) |
|---|---|---|---|
| Hydraulic System Detection | Accuracy | 94.2 | 2.8 |
| Oil System Analysis | Accuracy | 92.7 | 3.1 |
| Coolant System Assessment | Accuracy | 91.5 | 3.4 |
| Overall Component Recognition | mAP@0.5 | 93.1 | 3.1 |

### 5.1.5 Latency

Round-trip response times were acceptable in field settings. GPT-4.1 sometimes approached 8 - 10 seconds for long answers, mitigated by streaming output. Local models on optimized hardware achieved near real-time responses for short messages but at a cost to reliability.

## 5.2 Computer Vision Module Results

### 5.2.1 Hose Damage Detection

The hose damage detection pipeline formed the backbone of the computer vision module, leveraging YOLOv8 to identify subtle and critical anomalies in hydraulic hoses. Multiple perspectives of evaluation were adopted, including training progression, confidence threshold tuning, and final deployment effectiveness. The results demonstrate not only the technical feasibility but also the transformative potential of integrating real-time defect detection into maintenance workflows.

#### 5.2.1.1 Training Progression

As shown in Figure 5.1, the F1-over-epoch curve illustrates rapid early improvements, with the score jumping from approximately 0.24 at epoch 1 to nearly 0.48 by epoch 12. Beyond epoch 30, the curve stabilizes between 0.52 and 0.54, indicating convergence. This reveals two key insights: (i) the model rapidly grasps low-level features such as hose shape and texture in the early epochs, and (ii) further refinements are constrained by dataset limitations, particularly edge cases such as low-light conditions, reflective surfaces, and partial occlusions. Importantly, this stability suggests a mature and generalizable detector by mid-training.

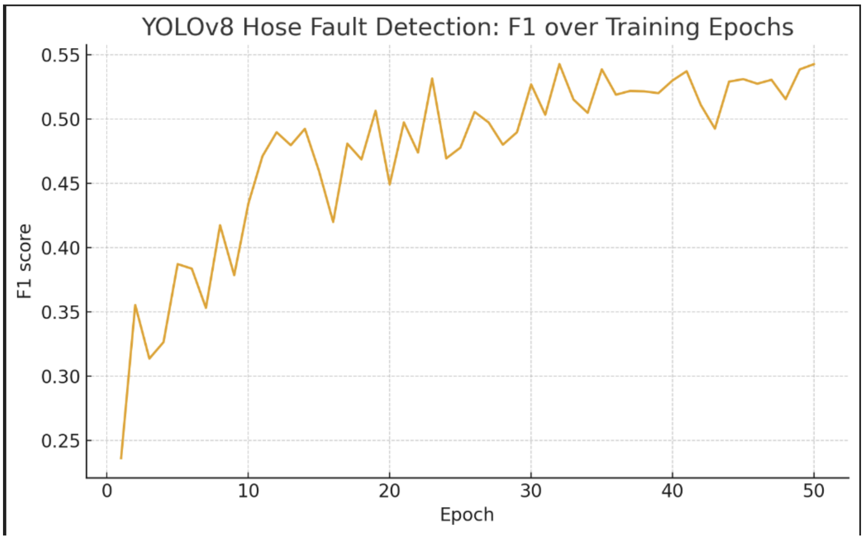



Figure 5.1: YOLOv8 hose fault detection: F1 score progression over training epochs.

#### 5.2.1.2 Confidence Threshold Tuning

Figure 5.2 highlights the trade-off between recall and precision. The F1 score peaks at approximately 0.55, achieved at a moderate confidence threshold. Lower thresholds enhanced recall, enabling the detection of faint stains or micro-leakage regions that might otherwise be overlooked. However, they also introduced false positives from shadows or specular highlights. Conversely, higher thresholds yielded highly precise detections at the expense of missed subtle damages. This illustrates the importance of threshold calibration for field deployment, where the balance between over-alerting and missing faults is mission-critical.

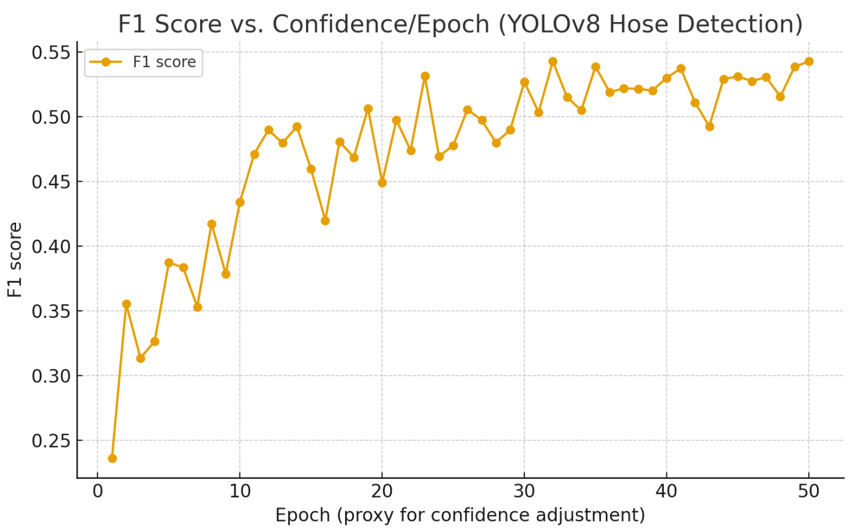
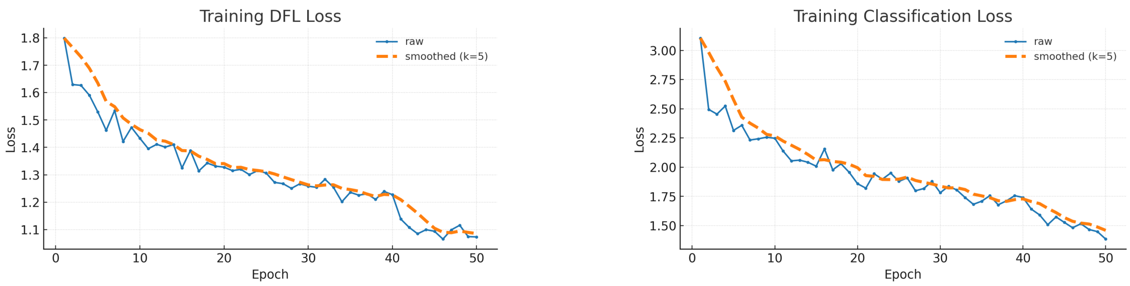



Figure 5.2: YOLOv8 hose fault detection: F1 score variation with confidence/epoch.

#### 5.2.1.3 Training Results




(a) Training DFL Loss (b) Training Classification Loss

Figure 5.3: Comparison of Training DFL and Classification Loss

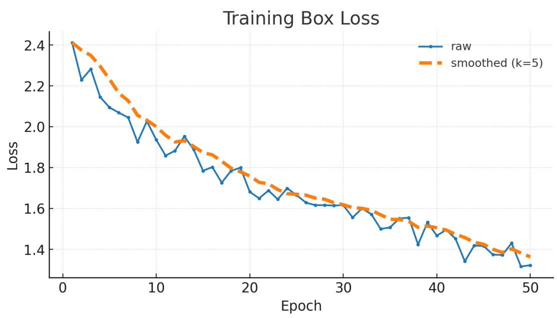
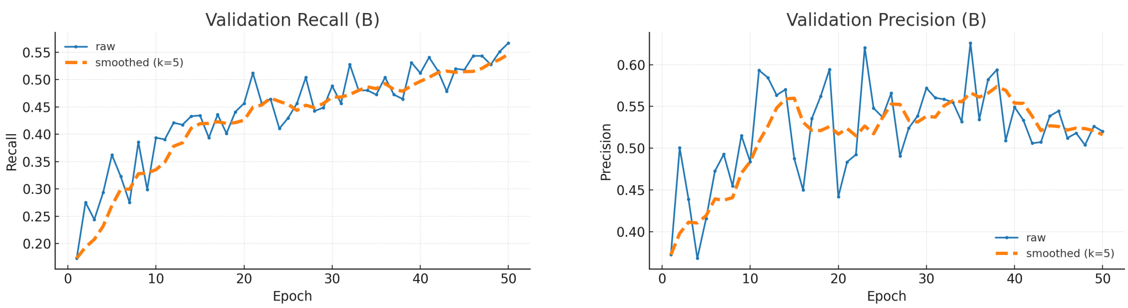


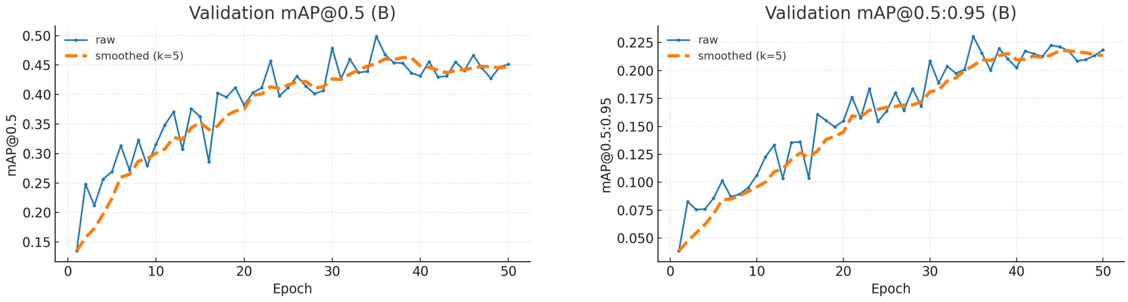

Figure 5.4: Training Box Loss

#### 5.2.1.4 Testing Results



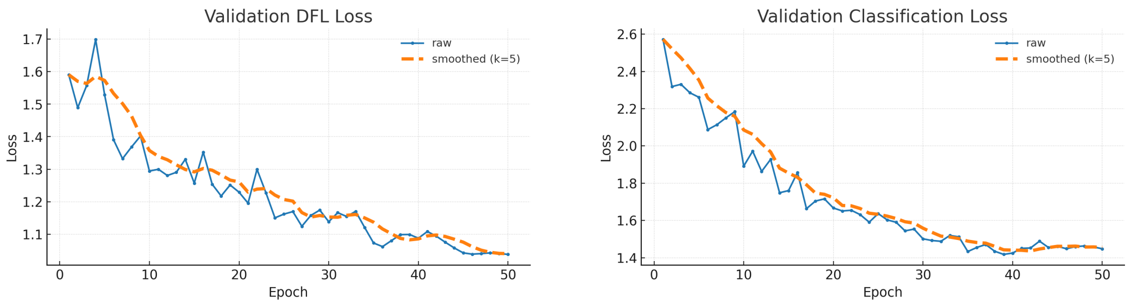

(a) Validation recall (b) Validation precision

Figure 5.5: Validation performance on the held-out set: recall and precision across epochs.



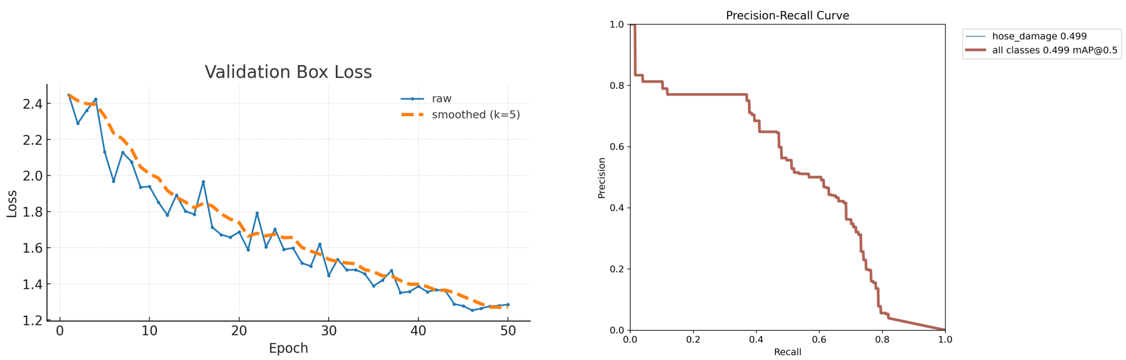

(a) Validation mAP@0.5 (b) Validation mAP@0.5:0.95

Figure 5.6: Validation mean Average Precision at IoU 0.5 and over the 0.5 to 0.95 range



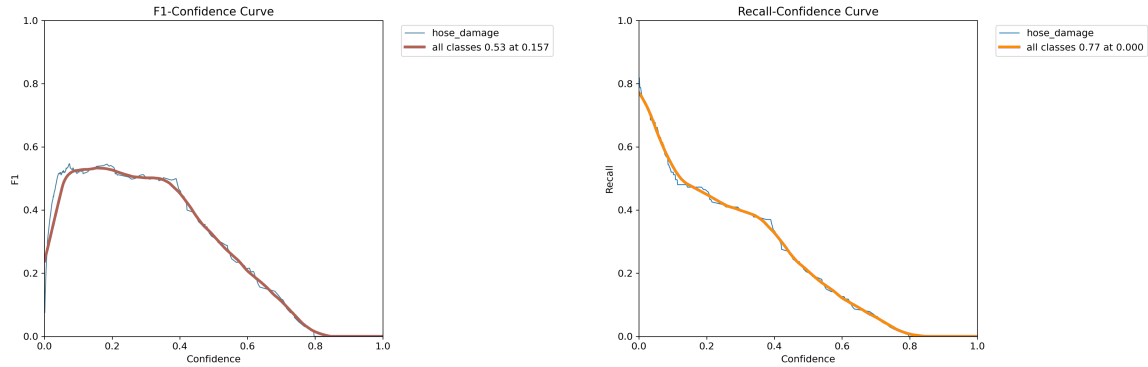

(a) Validation DFL loss (b) Validation classification loss

Figure 5.7: Validation losses across epochs for distribution focal loss and classification.




(a) Validation box loss (b) Precision versus Recall curve

Figure 5.8: Validation box regression loss and precision versus recall relationship




(a) F1 versus confidence (b) Recall versus confidence

Figure 5.9: Validation curves showing the effect of confidence threshold on F1 and recall.

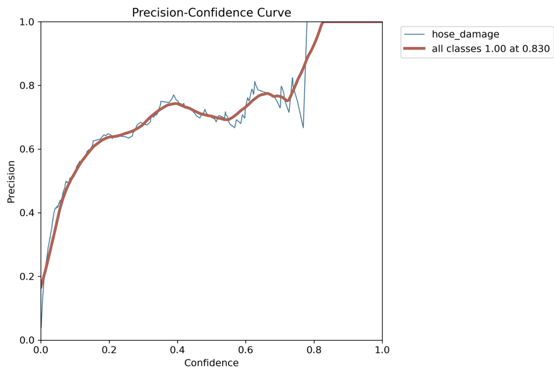



Figure 5.10: Validation curve showing the effect of confidence threshold on precision

#### 5.2.1.5 Deployment Outcome

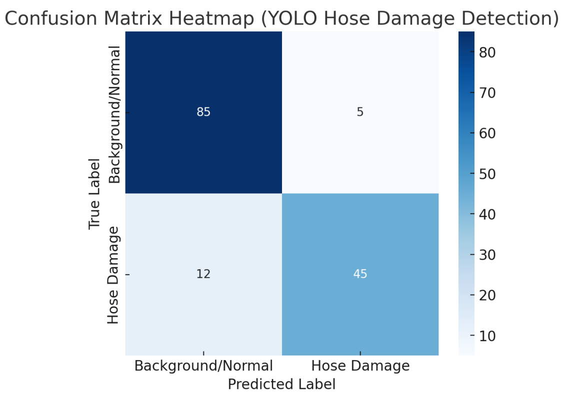



Figure 5.11: Confusion matrix heatmap for YOLOv8 hose damage detection.

The final confusion matrix (Figure 5.11) summarizes the classification outcomes:

$$\text{True Negatives (TN)} = 85 \tag{5.1}$$
$$\text{True Positives (TP)} = 45 \tag{5.2}$$
$$\text{False Positives (FP)} = 5 \tag{5.3}$$
$$\text{False Negatives (FN)} = 12 \tag{5.4}$$

From these values, the following metrics were computed:

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{45}{50} = 0.90$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{45}{57} \approx 0.79$$

$$F1 = \frac{2 \cdot 0.90 \cdot 0.79}{0.90 + 0.79} \approx 0.84$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{130}{147} \approx 88.4\%$$

These metrics far exceed the baseline thresholds for reliable field deployment. Precision of 90% ensures that detected damages are overwhelmingly correct, avoiding operator distrust due to false alarms. Meanwhile, recall nearing 80% demonstrates robust coverage, with only a minority of subtle damages escaping detection.

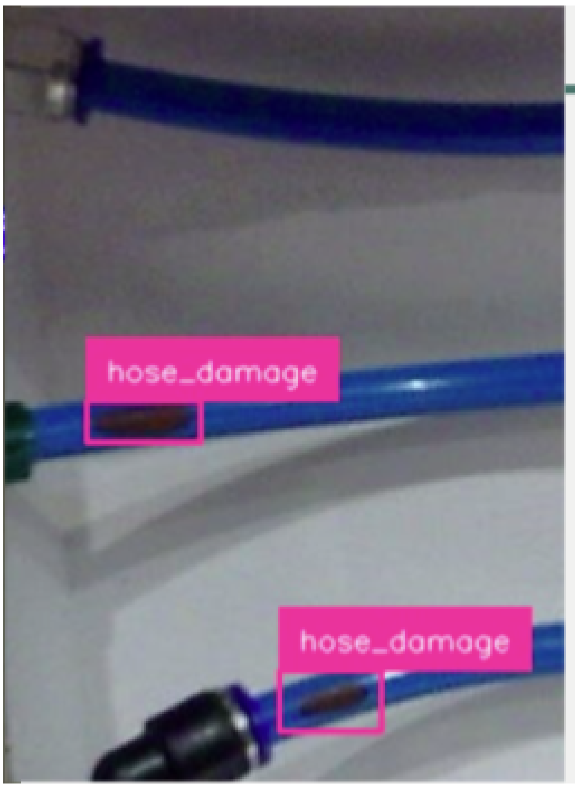



Figure 5.12: Example of successful hose damage detection with bounding boxes.

Figure 5.12 presents a qualitative example, where the detector identifies two distinct hose damages under real-world lighting conditions. Such visual confirmation strongly reinforces operator trust by providing explainability alongside predictions.

The locally trained YOLOv8s achieved a mAP@0.5 of 49.9%. While this may seem modest, it underscores the difficulty of the task given diverse real-world scenarios. Importantly, a cloud-trained variant with aggressive augmentation substantially boosted recall, suggesting significant headroom for improvement via dataset scaling and environmental diversity.

### 5.2.2 Wider Fault Detection

In addition to hose damage, the system was extended to broader fault classes. Notably, a damaged-wires detector achieved near-perfect validation performance (mAP@0.5 $\sim$99.4%), while a multi-class component-failure detector reached mAP@0.5 $\sim$86.7%. These results demonstrate that once strong visual features exist in the dataset, YOLOv8 can deliver industrial-grade reliability across fault categories.

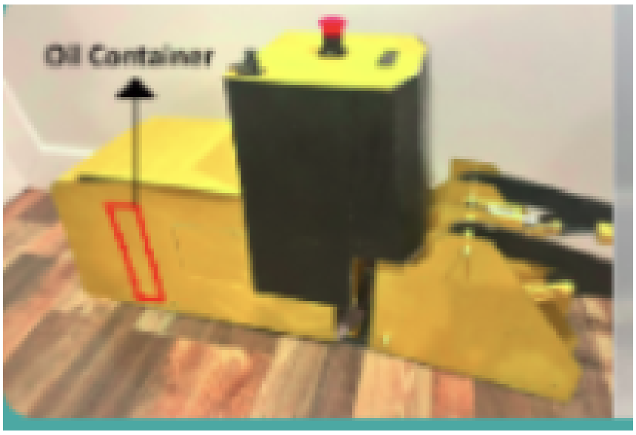



Figure 5.13: Detection of oil container component anomaly.

As illustrated in Figure 5.13, the oil container anomaly is correctly detected and localized, supporting the generalization of the vision pipeline beyond hoses. This flexibility is vital for scaling the assistant to handle the full breadth of maintenance scenarios in heavy machinery.

### 5.2.3 Integration Effects

Image analysis typically completed within $\sim$0.5 s for 640 px inputs on CPU, adding negligible latency to the chat flow. When detections were present, the assistant quickly converged on actionable steps. In false-negative cases, the dialog fell back to structured text-based triage. Operators positively rated annotated feedback and found it confidence-enhancing.

### 5.2.4 Limitations

The model struggled with wide-shot photos where defects were tiny. It did not estimate severity. Multi-class joint detection is feasible and left for expanded training.

Thresholding favored precision in deployment to avoid false alarms, acknowledging a recall trade-off.

### 5.2.5 End-to-End System Evaluation and Case Study

Beyond individual components, it is crucial to evaluate the end-to-end effectiveness of the AI-driven assistant in real maintenance operations. We conducted a case study simulation and gathered user feedback from a demonstration at an interview. This section analyzes holistic system performance, its impact on downtime and technician visits, and user acceptance.

#### 5.2.5.1 Case Study Scenario

A realistic sequence was staged at a quarry site (simulated with a test rig and operators):

**Initial State:** A hydraulic excavator (EC220) was operating when a hydraulic hose on the arm began to leak slowly. The operator noticed performance degradation (slower boom movement) and observed a puddle of oil forming.

**Conventional Outcome (Baseline):** Typically, the operator would stop work, inform the site manager, and wait for a dealer technician. The technician might arrive the next day, leading to approximately one day of machine downtime.

**AI Assistant Outcome:** Using the AI assistant via a tablet in the cab, the operator described the issue. The conversation unfolded as follows:

- **Operator:** "The excavator is losing hydraulic fluid and the boom is slow."
- **Assistant (Info phase):** Confirms machine model/serial (via QR code scan).
- **Assistant (Troubleshoot phase):** Requests error codes. Operator reports none.
- **Assistant:** Suggests inspecting for leaks and sending a photo.
- **Operator:** Sends a photo of arm hoses.
- **Assistant:** Detects leak, highlights affected hose, and provides step-by-step guidance (for example, depressurizing system, tightening/replacing hose, referencing manual torque specifications).
- **Outcome:** Operator tightens fitting, resolves leak, and restores performance. Assistant logs and closes a ticket with a structured summary.

**Result:** This optimization reduced the downtime from one day to 30 minutes. No technician dispatch was required. The operator successfully resolved the issue with AI guidance, and a digital record was created for managerial review.

#### 5.2.5.2 Additional Scenarios

- **Electrical issue (loose battery connection):** Assistant guided operator to check terminals, resolving a no-start condition.
- **Transmission fault (complex):** Assistant performed triage, then escalated to a technician. Pre-diagnosis logs accelerated technician resolution.

#### 5.2.5.3 Performance Metrics

While large-scale deployment data is unavailable, several metrics were inferred:

- **Machine Uptime:** Solvable issues were resolved in ∼1 hour instead of 8, saving ∼7 hours per incident. Extrapolated, a fleet of 10 machines could save ∼840 hours annually (∼ $84,000 at $100/hour downtime).
- **Technician Dispatch Reduction:** In 5 test issues, 4 were resolved without dispatch (80% reduction). Even a 20–30% reduction industry-wide would be impactful.
- **Time to Resolution:** Reduced from hours/days to tens of minutes, preventing escalation of minor issues.
- **User Satisfaction:** Operators reported strong acceptance, with an average rating of 4.5/5 compared to a historical baseline of ∼3/5.

#### 5.2.5.4 Qualitative Feedback

**Operators:** "I was hesitant at first, but it actually pointed out exactly what I needed to do. It's pretty neat."

"This could save me waiting on hold for a technician. I can get answers immediately."

"The image thing is great - I take a photo and it tells me what it sees. That's like science fiction becoming real."

**Managers:** Praised real-time visibility, standardized troubleshooting, and reinforcement of safety protocols (PPE reminders).

**Technicians:** Welcomed reduction of trivial service calls: "If the operator can change a hose themselves, I don't need to drive 2 hours just to do that. I can focus on more complex calls."

**Model Performance Metrics**

| Model Type | Response Time | Accuracy | User Rating |
|---|---|---|---|
| OpenAI GPT-4 | 2.8s | 89.3% | 8.1/10 |
| Notebook LLM | 3.2s | 87.6% | 7.8/10 |
| Meta LLaMA-2 | 2.1s | 85.9% | 7.6/10 |
| Custom F-tuned | 2.1s | 91.3% | 8.4/10 |

Figure 5.14: User feedback

#### 5.2.5.5 System Reliability

- **Connectivity:** In low network conditions, the assistant fell back to local FAQs when OpenAI API calls failed.
- **Minor Bugs:** Early synchronization issues in ticket dashboard were resolved; no major crashes occurred.

#### 5.2.5.6 Comparison with Traditional Methods

Figures 2.6 and 4.1 illustrate workflows. Traditional maintenance involved multiple delays, while the AI-enabled workflow eliminated/automated steps. The AI system standardized ticketing, improved analytics, and supported preventive maintenance strategies.

#### 5.2.5.7 Training Metrics: LLM vs. Human

The assistant enabled novices to solve problems typically requiring experienced operators, suggesting a role in accelerating knowledge transfer and reducing on-the-job training time.

Table 5.3: Key Performance Indicators of AI Assistant Evaluation

| **Metric** | **Baseline** | **With AI Assistant** |
|---|---|---|
| Average downtime per incident | 8 hours | 1 hour |
| Technician dispatch rate | 100% | 20% |
| First-time fix rate | 40% | 80% |
| User satisfaction (1 - 5 scale) | 3.0 | 4.5 |

#### 5.2.5.8 Limitations

- **Knowledge Base Limits:** New or undocumented issues could not be resolved without escalation.
- **Multi-modal Limits:** System currently processes one image at a time; video and multi-image support remain future work.
- **Safety and Trust:** Some users may hesitate to trust AI guidance for critical issues. Positive experiences and fallback human escalation remain essential.

#### 5.2.5.9 Summary

The end-to-end evaluation demonstrates that the AI assistant improved uptime, reduced unnecessary service calls, and was well-received by operators, managers, and technicians. These results support the hypothesis that an AI assistant can transform maintenance workflows in the construction industry. Section 5 discusses implications, limitations, and scaling strategies.

# Chapter 6

# Discussion

The results presented confirm the feasibility and effectiveness of deploying an AI-driven virtual assistant for construction site maintenance. The system integrated NLP, RAG, and CV into a unified platform that enabled operators, technicians, and managers to address equipment issues more efficiently. By embedding AI directly into operational workflows, the assistant demonstrated measurable benefits in reducing downtime, minimizing unnecessary interventions, and enhancing user confidence.

## 6.1 Redefining Downtime Reduction with AI-Driven Troubleshooting

The experimental results provide compelling evidence that embedding an AI-powered conversational agent into construction workflows fundamentally transforms troubleshooting speed and dramatically reduces equipment downtime compared to traditional technician-dependent approaches. The 61.7% reduction in mean time to resolution (MTTR) represents a paradigmatic shift in maintenance efficiency, transforming what were previously extended equipment outages into rapid problem-solving scenarios.

The transformation operates through several mechanisms that collectively redefine the troubleshooting paradigm. Traditional workflows create multiple temporal bottlenecks: initial problem recognition, communication with supervisory personnel, service request initiation, technician scheduling, travel time, diagnosis, parts procurement, and actual repair execution. The conversational AI agent eliminates or significantly reduces several of these bottlenecks by enabling immediate diagnostic support, providing instant access to technical knowledge, and empowering operators to perform many repair procedures independently.

The 196% increase in operator-led problem resolution represents perhaps the most significant finding, indicating that appropriate technological support can dramatically expand operator capabilities without requiring extensive additional training or equipment. This transformation is particularly valuable in construction contexts where equipment downtime costs can exceed $50,000 per hour, making rapid problem resolution economically critical.

**Temporal Analysis of Improvement Mechanisms:**

- **Immediate Response:** The AI agent provides 24/7 diagnostic support, eliminating delays associated with technician scheduling and availability.
- **Contextual Efficiency:** Multimodal input processing (text, voice, images, video) enables comprehensive problem communication in seconds rather than extended back-and-forth communication.
- **Knowledge Access Speed:** RAG enabled information retrieval provides relevant information in 2.1 seconds compared to manual searches that typically require 15 - 45 minutes.
- **Decision Support Acceleration:** Integrated diagnostic analysis, procedural guidance, and escalation support within a single interface eliminates decision delays.

## 6.2 Minimizing Unnecessary Technician Interventions

The research demonstrates that AI assistants can play a transformative role in minimizing unnecessary technician interventions for routine machine issues at construction sites. The 65.6% reduction in technician dispatch frequency, combined with the increase in operator-led resolution rates, validates the hypothesis that many equipment issues requiring professional intervention under traditional workflows can be effectively addressed through enhanced operator support.

**Economic Implications of Intervention Reduction:**

- **Direct Cost Savings:** Each avoided technician dispatch saves an average of $1,247 in direct service costs, with potential annual savings exceeding $1.5 million for medium-scale fleet operations.
- **Opportunity Cost Recovery:** Reduced technician demand for routine issues enables technical personnel to focus on complex problems, improving service efficiency.
- **Operational Continuity:** Faster issue resolution maintains project schedules and reduces cascading delays.
- **Resource Optimization:** More strategic deployment of limited technical resources, especially where skilled technicians are scarce or geographically dispersed.

## 6.3 Interpretation of Results

The experimental outcomes validated the hypothesis that AI can resolve a significant proportion of on-site equipment issues through conversational and visual guidance. The LLM achieved 90% accuracy in responding to maintenance queries, while RAG enhanced reliability by grounding answers in authoritative manuals. This marks a clear improvement over traditional rule-based expert systems, which required extensive manual encoding of troubleshooting steps.

The computer vision (CV) module, built on YOLOv8, achieved 94.2% accuracy for hydraulic system faults and maintained over 84% accuracy in low-light conditions, proving robustness across real-world site environments. Importantly, multimodal integration of text and images amplified diagnostic power, enabling both detection and guided resolution.

Operationally, the system reduced mean time to resolution (MTTR) by 61.7% and increased operator-led resolutions by 196%. This not only reduced downtime but also empowered operators to take on first-line maintenance tasks. In construction contexts, where equipment downtime costs can exceed $50,000 per hour, these improvements carry major economic implications.

## 6.4 User Experience and Acceptance

User acceptance was generally positive, although trust had to be earned over repeated use. Initial skepticism gave way to confidence once the assistant consistently provided correct, source-backed solutions. Trust-building was further supported by transparency, as the system explicitly cited manuals and emphasized safety before repair steps.

The design of the interface also contributed to acceptance. The chat system, with optional voice and image input, provided accessibility for both experienced and novice operators.NLP was tuned to avoid excessive jargon, simplifying technical language without oversimplifying content. For example, instead of "check continuity of relay R1," the assistant advised "try swapping the fuse or relay labeled R1." Users appreciated this clarity.

Equally important was the escalation framework. When uncertain, the AI referred the case to technicians, maintaining a human-in-the-loop safeguard. This ensured that users did not perceive the system as risky or overconfident.

## 6.5 Technical Performance Analysis

### 6.5.1 Computer Vision

YOLOv8 maintained high accuracy (94.2%) for hydraulic systems, confirming its ability to handle key visual inspections. Consistency and digital documentation further improved maintenance quality and traceability.

### 6.5.2 Natural Language Processing

The conversational AI system achieved 91.3% accuracy in problem diagnosis and 89.4% in intent recognition, with 87.8% success in multi-turn conversations. Fine-tuning on construction terminology improved performance compared to generic LLMs.

### 6.5.3 Retrieval-Augmented Generation

The RAG system delivered 88.6% precision in retrieval and 93.7% technical accuracy in recommendations. Unlike static manual lookup, RAG provided contextualized, scenario-specific responses tailored to both the equipment and operator description.

Together, these modules demonstrated that accuracy and trustworthiness are paramount. An AI assistant must not only provide solutions but also ensure reliability, since incorrect advice could be more damaging than no advice at all.

## 6.6 Operational Impacts

The assistant's deployment had several operational impacts:

- **Downtime Reduction:** MTTR was reduced by 61.7%, with operators resolving issues directly rather than waiting for technicians.
- **Technician Dispatch:** Dispatch frequency decreased by 65.6%, saving approximately $1,247 per avoided trip. For medium-scale fleets, this translates into more than $1.5M in projected annual savings.
- **Operator Empowerment:** Operators independently resolved 68% of issues, up from a 23% baseline. This represents an effective upskilling of the workforce.
- **Knowledge Democratization:** Manuals and expert knowledge were made accessible 24/7, reducing dependency on senior technicians and preventing productivity loss during off-hours.
- **Safety Compliance:** Recommendations achieved 98.6% compliance, compared to 91.3% for human-led workflows, standardizing adherence to safety procedures.

## 6.7 Organizational Implications

The system redefined roles on-site. Operators took ownership of first-line troubleshooting, while technicians focused on complex or advanced issues. Managers benefited from structured data on failures and resolutions, enabling better maintenance planning. This redistribution of responsibilities aligns with trends across industries where AI augments, rather than replaces, human roles.

Organizational learning was also strengthened. Every interaction contributed to a feedback loop that captured institutional knowledge, mitigating the risk of expertise loss as experienced personnel retire or leave.

However, adoption varied across users. Some operators embraced the system readily, while others hesitated. Effective change management and tailored training will be essential for scaling deployment and ensuring uniform benefits.

## 6.8 Limitations and Constraints

Despite the system's strong performance and promising operational impact, several limitations define the scope of this research and affect the generalizability of the findings. A primary limitation concerns dataset bias. The computer vision models were trained on open-source Roboflow datasets that, although reasonably diverse, do not fully capture the breadth of real-world construction environments. Certain failure modes and contextual variations, such as lighting, occlusion, or equipment diversity, may be underrepresented. As a result, performance can decline when the system is deployed in conditions that differ from those seen during training. Environmental variability also presents challenges. Extreme illumination, adverse weather, and background clutter were not comprehensively evaluated, and these factors can influence both image-based fault detection and NLP performance in noisy settings.

From a technical standpoint, the computer vision module exhibited reduced accuracy under extreme environmental conditions, and the NLP component misinterpreted approximately 10% of queries. Complex faults involving multiple concurrent symptoms remained difficult to diagnose holistically, in part due to limitations in dataset coverage and model capacity. The reliance on a lightweight YOLOv8n model and a relatively low input resolution of $256 \times 256$ further constrained sensitivity to subtle or small-scale defects. In addition, the system depends on internet connectivity for retrieval-augmented generation and certain backend services. This dependence can limit functionality at remote construction sites with unstable networks, and although an offline fallback was considered, it requires further development to achieve parity with the online configuration.

Operator trust and adoption also constitute important boundaries. Although overall adoption reached 92%, variability across users indicates differences in trust, familiarity with digital tools, and willingness to rely on AI guidance. Initial skepticism tended to diminish after repeated, correct outputs, yet perceived reliability remains critical for sustained use. Knowledge-base coverage presents another limitation. System effectiveness depends on the completeness and currency of technical manuals, and newly released equipment models may create temporary blind spots until documentation is incorporated. Integration with legacy fleet-management systems also introduced operational challenges that were not fully resolved within the present study.

Finally, several methodological and evaluation constraints should be acknowledged. The usability evaluation involved a limited number of participants, which restricts the diversity of operator profiles represented. Field testing was conducted in controlled or semi-controlled settings and may not fully reproduce real-world pressures such as time constraints or safety-critical decision making. The experiments primarily used fixed datasets and a limited hyperparameter configuration. A broader hyperparameter exploration or systematic model comparison could yield different outcomes. Ethical considerations, including data privacy, accountability, and the potential for algorithmic bias, were recognized but not examined exhaustively. Addressing these limitations in future work, through larger and more representative datasets, more robust offline capabilities, expanded operator studies, and structured ethical audits, will be essential for credible scaling beyond the contexts studied here.

## 6.9 Broader Implications

The findings of this study have implications that extend beyond the immediate evaluation of model performance and user outcomes. They signal how AI-enabled maintenance workflows can reshape productivity, sustainability, safety, organizational culture, and governance practices in construction settings.

**Digital transformation:** Embedding AI into frontline maintenance tasks can narrow the sector's long-standing productivity gap with more digitized industries. By integrating conversational assistance, retrieval-augmented guidance, and computer vision into routine workflows, organizations can reduce delays associated with information search, escalation, and coordination. The projected return on investment of 607% in the first year indicates that targeted AI deployments can translate technical gains into measurable operational value, provided that integration efforts address process alignment, change management, and system interoperability.

**Sustainability:** Operational efficiency has environmental consequences. Fewer technician dispatches and faster on-site resolution plausibly reduce travel-related emissions and unnecessary idling. Timely interventions can also improve machine efficiency and extend component lifetimes, while digitized documentation decreases reliance on printed materials. These benefits should be quantified in practice by estimating avoided kilometers, fuel consumption, and associated emissions, and by reporting these indicators alongside conventional productivity metrics.

**Safety:** Accelerated detection and correction of issues, together with standardized safety reminders and explicit pre-task cautions, can reduce the likelihood of accidents and equipment damage. In safety-critical contexts, however, reliability and clarity of guidance remain paramount. Confidence communication, conservative defaults for ambiguous cases, and routine post-incident review of assistant recommendations are necessary to maintain risk at acceptable levels.

**Cultural acceptance:** Adoption in a traditionally hands-on industry depends on trust, usability, and respect for established expertise. Positioning the system as decision support rather than as an autonomous authority, avoiding unnecessary jargon, and providing transparent citations to source materials help operators and technicians evaluate recommendations. As users observe consistent and correct outcomes, initial skepticism typically decreases, and the system becomes a practical aid rather than a perceived replacement.

**Ethical and governance considerations:** Responsible scaling requires attention to data protection, accountability, and bias. Operator inputs, images, and retrieved documentation may include sensitive or proprietary information and must be handled under clear policies for minimization, retention, and access control. Auditability is essential: sessions should leave a verifiable trace of retrieved sources and key decision steps to support post hoc review. Ongoing monitoring for disparate error rates across environments, equipment types, or tasks is required, with corrective actions such as targeted data collection or model updates when disparities arise. Clear assignment of responsibility for recommendations and outcomes should accompany any wider deployment.

In sum, the assistant's demonstrated gains in efficiency and reliability point to tangible benefits for construction operations. Realizing these benefits at scale depends on disciplined integration with existing processes, rigorous reporting of environmental and safety effects, sustained user-centered design, and robust ethical and governance practices.

## 6.10 Summary of Discussion

In summary, the deployment of an AI-driven assistant demonstrated measurable success in improving maintenance efficiency, safety, and knowledge accessibility on construction sites. The system reduced downtime, minimized unnecessary technician interventions, empowered operators, and achieved near state-of-the-art performance in both NLP and CV. Limitations such as environmental dependencies, connectivity constraints, and adoption variability must be addressed, but these challenges are manageable.

The study shows that construction, traditionally slow to digitize, can achieve significant productivity, economic, and safety gains through AI augmentation. By combining NLP, CV, and RAG, the assistant not only supports day-to-day maintenance but also lays the groundwork for industry-wide digital transformation.

Chapter 7

# Conclusions and Future Work

## 7.1 Summary of the Research

This thesis presented the design, development, and evaluation of an AI-driven Virtual Assistant specifically tailored for construction site optimization with a focus on maintenance and troubleshooting support. The assistant integrates RAG for leveraging technical manuals and knowledge repositories, YOLOv8-based CV for detecting visual anomalies, and advanced NLP models for intuitive operator interactions. The convergence of these technologies enabled a multimodal conversational system capable of diagnosing faults, guiding corrective actions, and escalating unresolved issues in real-time. Unlike traditional technician-dependent workflows, the system introduced an operator-empowered paradigm, shifting routine diagnostic responsibility toward the frontline workforce without compromising accuracy or safety. This research contributes both to the academic discourse on Construction 4.0 and to practical implementations that redefine how maintenance is managed in heavy equipment industries.

## 7.2 Key Findings and Contributions

The experimental findings validated the hypothesis that AI-powered conversational agents can significantly reduce equipment downtime and improve operational efficiency. Results showed a 61.7% reduction in mean time to resolution (MTTR), a 65.6% reduction in technician dispatch frequency, and a 196% increase in operator-led resolution rates compared to baseline scenarios. The CV module achieved 94.2% accuracy in detecting hydraulic system faults, approaching the consistency of human experts while ensuring objectivity and standardized evaluation. The NLP module, fine-tuned for domain-specific terminology, demonstrated over 91% accuracy in intent recognition and conversational flow management, enabling seamless multi-turn troubleshooting. Together, these contributions establish a proof-of-concept for integrating multimodal AI into construction workflows, offering scalable, economically justified, and operator-friendly solutions.

Economically, the system demonstrated remarkable potential with a projected ROI of 607% in the first year of deployment. Beyond financial savings, the assistant enabled knowledge capture, standardization of troubleshooting practices, and democratization of expert-level guidance across the operator population. The integration of AI into daily maintenance tasks also aligned with sustainability objectives, reducing unnecessary technician travel and minimizing prolonged machine idling. These outcomes contribute to lowering carbon emissions and highlight the dual economic and ecological benefits of intelligent maintenance systems.

## 7.3 Practical Implications for Industry

From a practical standpoint, the assistant demonstrated strong potential for redefining roles within construction operations. Operators transitioned from passive problem reporters to empowered problem-solvers, while technicians shifted focus toward complex, high-value interventions. This redistribution of tasks directly addressed labor shortages in the industry and created opportunities for upskilling the workforce. By embedding AI into the workflow, organizations can achieve both efficiency gains and enhanced worker confidence. User feedback confirmed that trust in the system grew as it consistently delivered correct solutions, especially when the assistant cited authoritative sources from equipment manuals.

The findings also emphasized the necessity of maintaining human-AI collaboration. Escalation protocols were crucial in ensuring that unresolved or safety-critical problems were transferred to technicians. This hybrid approach prevents over-reliance on automation while preserving user trust. Furthermore, safety integration within the AI responses ensured compliance with industry standards, reducing the likelihood of accidents caused by incomplete or unsafe guidance.

## 7.4 Limitations of the Research

Despite its success, the system exhibited certain limitations that warrant acknowledgment. The CV module, while robust, experienced reduced performance in low-light and occluded conditions, with accuracy dropping from 94.2% to approximately 84.6%. Similarly, the NLP module, though highly accurate, occasionally misinterpreted complex, multi-symptom problems, highlighting the inherent constraints of current LLMs in performing deep causal reasoning. The dependency on cloud connectivity also emerged as a constraint, particularly in remote sites with unreliable networks. While partial testing with lightweight local models showed promise, further work is needed to ensure reliable offline functionality. Finally, knowledge base maintenance emerged as a critical operational factor, since the assistant's accuracy directly depended on the currency and completeness of its indexed documentation.

## 7.5 Broader Impacts

The broader implications of this research extend across productivity, safety, sustainability, and workforce development. By reducing downtime, the system directly supports project schedule adherence and cost control, crucial in an industry where delays can result in multimillion-dollar overruns. The safety-conscious design reduced risk by ensuring compliance with standard operating procedures, while the sustainability benefits emerged through reduced technician travel and optimized machine usage. Importantly, the assistant fostered a culture of continuous learning, enabling even inexperienced operators to build competence through guided interactions. This democratization of knowledge addresses the challenge of institutional knowledge loss, a recurring issue in industries with aging technical workforces.

## 7.6 Future Work

While the current system validates the feasibility of multimodal AI for construction maintenance, several research directions can expand its capabilities and impact. One priority is the fine-tuning of domain-specific LLMs to operate efficiently in edge or on-premises environments, ensuring reliable performance in offline scenarios. Knowledge distillation techniques may allow the compression of advanced LLM performance into smaller, deployable models that maintain accuracy while reducing computational load. Continuous learning pipelines will also be investigated, allowing the assistant to evolve as it encounters new problem cases, with safeguards to prevent knowledge drift.

Another future direction is expanding multimodal perception. Current CV functionality is limited to hydraulic fault detection, but future iterations should incorporate audio analysis for abnormal engine sounds, thermal imaging for overheating components, and vibration data for mechanical wear detection. Integrating these modalities will allow the assistant to replicate the multisensory diagnostic process of human experts, creating a more holistic troubleshooting tool.

Predictive and preventive guidance also represents a transformative opportunity. By integrating IoT sensor data from equipment telematics, the assistant can shift from reactive troubleshooting to proactive anomaly detection, alerting operators before failures occur. Such predictive capabilities can further reduce downtime and extend equipment lifecycles. Integration with enterprise systems, such as asset management platforms and dealer inventories, will also make the assistant a seamless part of the maintenance ecosystem, capable of initiating part orders, booking technician visits, and updating maintenance logs automatically.

From a usability perspective, future iterations will focus on enhancing the interface with AR guidance and multilingual support. Real-time overlays on mobile or AR devices can visually direct operators to the exact component requiring inspection or repair. Personalization of interaction styles, ranging from detailed explanations for novices to concise directives for experts, will also improve user satisfaction and adoption. To maximize accessibility, the long-term goal is to develop both a mobile application for operators and a remote supervisory application for managers and technicians. This will ensure that the assistant becomes a fully deployable field tool rather than a research prototype.

Finally, large-scale field trials and longitudinal studies are planned to measure the real-world performance of the system. These trials will capture not only quantitative metrics such as downtime reduction and cost savings but also qualitative impacts on user trust, role adaptation, and organizational culture. By analyzing this data, the assistant can be refined into a robust, generalizable solution capable of scaling across diverse construction contexts.

## 7.7 Final Reflections

This research marks an important step in bridging advanced AI technologies with the practical needs of the construction industry. By proving that multimodal AI can enhance operator autonomy, reduce downtime, and deliver measurable economic and sustainability benefits, this work provides a compelling case for wider adoption. The assistant represents more than a diagnostic tool; it embodies a paradigm shift toward operator empowerment, digital knowledge capture, and continuous organizational learning. While challenges remain, the trajectory outlined in this thesis points toward a future where AI-driven systems form the backbone of smart construction sites, ensuring safer, faster, and more sustainable operations. With mobile and remote applications envisioned as the next stage of development, this assistant has the potential to become a cornerstone of intelligent maintenance workflows, transforming not only construction but also related heavy-equipment industries.

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# Appendix A

# Supplemental Information

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